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"Do extra-sentential distractors lead to retrieval interference? A simulated ERP study"

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Abstract

This thesis provides the theoretical framework for a study that investigates the effects of sentence-external distractors on sentence processing with a specific focus on their influence on the N400 amplitude in EEG data. Using a simulated EEG study, it was investigated how external memory items can influence sentence comprehension by specifically focusing on the semantic interference effects of sentence-external memory items. The study further aims to differentiate between encoding and retrieval interference by analysing the regions in which an effect can be observed. The results indicate that the N400 amplitude is largest when the external distractor items semantically match the cues of the verb. A Bayesian analysis confirms the existence of an interaction between semantic interference and memory load which supports the hypothesis that sentence-external memory items can influence sentence processing. The findings seem to point towards a retrieval interference effect during memory processing since the interference effect has been shown to appear in the critical region. This supports the theories of cue-based retrieval interference and indicates that a semantic match between sentence-external memory items and the target item can influence the retrieval difficulty.

1. Introduction

In human languages, the ability to process sentences is of high importance for making sense of the languages that surround us. Comprehending a sentence includes establishing syntactic connections between sentence constituents to be able to completely understand the meaning of a sentence. Since in human languages not all dependent grammatical connections are directly adjacent to each other, it seems that effective sentence comprehension relies on the ability of retaining grammatical connections in memory until they can be integrated into the broader sentence context. This retention of grammatical connections is based on memory processes that also include the working memory (Lewis & Vasishth, 2005) and seem to have the ability to recall previously encountered linguistic material. It has been shown that increased memory load hinders the retrieval of those grammatical connections (Schoknecht et al., 2023; Van Dyke & McElree, 2006). Therefore, for a better understanding on how sentence processing functions, it is also necessary to investigate the connection between memory and sentence processing to be able to draw conclusions on how memory processes might affect sentence comprehension. When investigating those memory processes, it is important to consider the different sentence processing mechanisms that could contribute to potential misinterpretations in sentence comprehension. A central question regarding those sentence processing mechanisms is whether the memory mechanisms that are involved take place during the encoding or retrieval stage of sentence processing and whether they involve proactive or retroactive interference.

Several different experimental paradigms have been used to investigate the interaction between memory and sentence processing and have found partially contrasting results (Mertzen et al., 2024; Schoknecht et al., 2023; Van Dyke & McElree, 2006). This thesis aims to contribute a framework to the ongoing debate by proposing a theoretical foundation for a study that further investigates the influence of memory on sentence processing by using a different experimental paradigm than the previous studies. It is specifically interested in examining the influence that sentence-external linguistic stimuli can have on sentence processing.

2. Background

2.1 Electroencephalography

A fundamental aspect of brain activity is the neuronal communication which is communication that is regulated by electrically active cells that are called neurons. The activity generated by neurons is called bioelectric activity. There exist several types of neurons among which pyramidal cells are the most important when it comes to measuring the bioelectric activity in the brain. Pyramidal cells can form complex networks which are connected through structures called dendrites with their dendritic spines and those are essential for synaptic transmission in the brain (Araya, 2014). Pyramidal cells show polarity which means that if one end of a dendrite has a positive charge, the other end automatically has a negative charge. The polarity of the dendrite is influenced by whether the received stimulus was inhibitory or excitatory and by the location of the synapse which can be either proximal or distal (Beres, 2017). Proximal synapses are located near the base of the dendrites, close to the cell body whereas distal synapses are further away, towards the dendritic tips. Exhibitory signals increase the firing potential of synapses and inhibitory signals decrease the firing, both processes also use different types of neurotransmitters (Purves & Williams, 2001).

To illustrate this with an example: when an excitatory stimulus reaches the distal end of a dendrite, the permeability of the cell membrane changes. This allows sodium ions (Na+) to enter the cell which creates a negative charge in the extracellular space. As a result, the proximal end of the dendrite becomes positive because the two ends of dendrites of a pyramidal cell always have opposite charges (Schomer & Da Lopes Silva, 2017). When a large number of pyramidal cells in an area undergo this process simultaneously, an electrical field is being created. The polarity of the extracellular space and thus the electrical field depends on various factors (see Schomer and Da Lopes Silva (2017) for a more detailed explanation) and the change is generally initiated by excitatory or inhibitory inputs from other neurons (Schomer & Da Lopes Silva, 2017). If the extracellular space becomes negative such as in the example above, the bioelectric activity that renders it negative can be measured by a non-invasive method called electroencephalography (EEG). EEG curves can be measured by placing recording electrodes at specific locations on the scalp and amplifying the voltage differences between them. In the case above, the

EEG-electrode will record a negative potential. The summed bioelectrical activity of pyramidal cells over time usually is referred to as postsynaptic potentials (Beres, 2017; Kutas & Federmeier, 2000). Postsynaptic potentials show slower changes in membrane potentials than other types of potentials (such as action potentials) and can last up to 200 milliseconds (ms) which makes it possible to measure their effects (Friederici & Thierry, 2008; Schomer & Da Lopes Silva, 2017).

Those measurable changes in the potential of the electric field can be elicited by a sensory, motoric or cognitive event and are referred to as event-related potentials (ERPs). When participants process external stimuli and respond to them, ERPs can be measured. These ERPs are small voltage fluctuations which are recorded through the placement of electrodes on the scalp. ERPs result from the summation of synchronised synaptic potentials following stimulus presentation. The resulting waveform fluctuates continuously and is divided into distinct components, characterised by their polarity (negative or positive deflections), scalp distribution, and latency (van Petten & Luka, 2006). A single ERP is too small to be seen but the ERPs of a specific event can be extracted by averaging multiple occurrences of similar events. These averaged occurrences then result in one voltage waveform at each recording site (each electrode) (Kutas & Federmeier, 2000).

The main advantage of using EEG in psycholinguistics is its exceptional temporal resolution that allows for tracking the brain activity at the millisecond or even sub-millisecond level. However, the spatial resolution of EEG is limited because some head tissues conduct electrical activity better than others which might cause the electrical current to disperse across the scalp. This makes it challenging to precisely localise the source of brain activity. Therefore, most ERP components in psycholinguistics are named after their time of occurrence rather than the place of occurrence (van Petten & Luka, 2006). In the following section one ERP component will be explained in more detail.

2.2 The N400

The N400 is an ERP component that is based on a centro-parietally distributed negativity and can be observed around 400 milliseconds after the presentation of a stimulus such as a word or picture. The name of this ERP component is based on its polarity (negative, therefore N) and its time of occurrence (roughly 400ms after stimulus onset). It has been shown to vary systematically with the processing of semantic information (Kutas & Federmeier, 2000). It is associated with semantic incongruities or difficulties in integrating a word into its given context. It was first identified by Kutas and Hillyard (1980), in a study that focussed on language processing of semantically anomalous information. Participants were presented with sentences that were either semantically congruent or incongruent. For instance, the sentence "He spread the warm bread with socks" (Kutas & Hillyard, 1984, p. 203) elicited a larger N400 response compared to the more contextually appropriate "He spread the warm bread with butter." The authors concluded that the N400 is particularly visible when semantically unexpected endings to sentences, such as “socks”, appear. Besides, the N400 is sensitive to long-term semantic relationships. For instance, an unexpected ending to a sentence that is categorically related to the expected ending elicits a smaller N400 compared to a categorically unrelated ending. The N400 also demonstrates sensitivity to masked lexical priming (Friederici & Thierry, 2008).

The amplitude of the N400 depends on several factors. Firstly, it can be influenced by how easily a word or concept fits into its preceding context, with the amplitude increasing with unexpectedness, unfamiliarity, or lack of match to the semantic context. Furthermore, as discussed by Kutas and Federmeier (2011) the amplitude of the N400 is influenced by semantic priming. For example, the amplitude of a content word is decreasing with each additional word in the sentence because the context becomes more established. Words that are highly expected elicit a small N400, whereas unexpected but still plausible words evoke a higher N400 and implausible words invoke the highest N400. An example can be seen in Figure 1.

The reduced N400 amplitude when the target word aligns with its the context indicates that prior information has created an expectation for upcoming words. The integration of this expected word seems to be easier since the expectation allows the semantic system

to prepare in advance for the upcoming word (Kutas & Federmeier, 2011). The reduction in N400 amplitude is inversely proportional to the cloze probability of the word: the more predictable a word, the smaller its N400 amplitude (Kutas & Hillyard, 1984). Additionally, highly frequent content words, words with more lexical neighbours, repeated words (Petten et al., 1991) and concrete words (van Petten & Luka, 2006) generally produce smaller N400 amplitudes (Kutas & Federmeier, 2000). This effect is observed in responses to both words and images that confirm or diverge from semantic expectations (Friederici & Thierry, 2008).

Figure 1 shows an example of how an N400 for expected vs. unexpected words may appear in the EEG. Here, ERPs that were recorded at the right medial central electrode can be seen. The participants read a sentence that was completed with an expected ("palms") or unexpected word ("pines" or "tulips"). Both, for *pines* as well as *tulips*, increased negativity compared to the target *palms* can be seen 250-500ms post-stimulus onset, classifying this as an N400. The reason why the two unexpected words show differences regarding their amplitude can be explained by the semantic difference between them. While p*ines* presents a violation for the context of the sentence, it is semantically very similar to the intended target *palms* whereas *tulips* presents a violation of the semantic context and comes from a different semantic category (flowers and not trees) (Kutas & Federmeier, 2000).

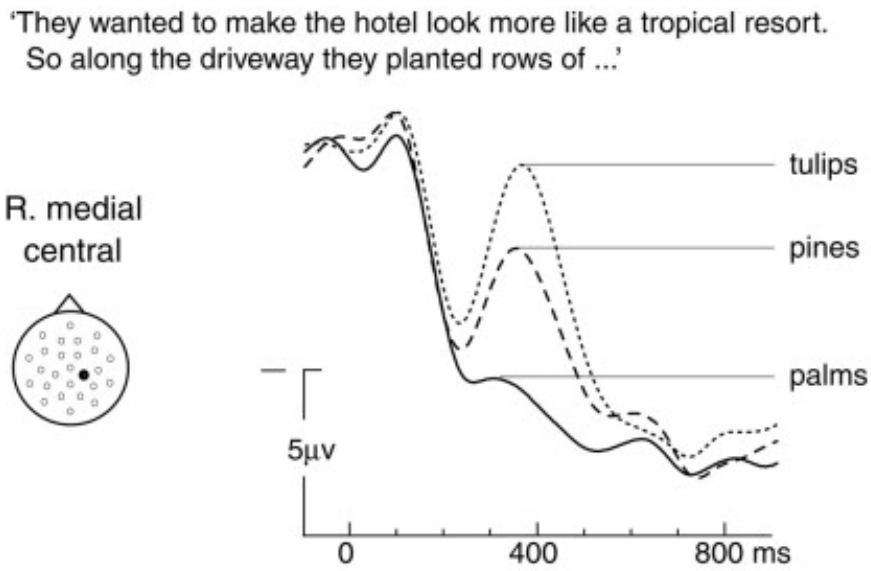

Figure 1. N400 Response to Semantic Plausibility, taken from Kutas and Federmeier (2000).

To measure how good a word fits into a given context, plausibility judgments from participants are an effective tool because they capture the subjective assessment of participants (Kutas & Federmeier, 2000). They are closely associated with the concept of predictive processing in language comprehension which assumes that during reading or listening, expectations about upcoming words are formed, and the alignment of these words or phrases with the surrounding context is continuously evaluated. Plausible continuations are processed more efficiently, whereas implausible continuations require increased cognitive resources (Kutas & Hillyard, 1984). Therefore, a correlation between N400 amplitude and plausibility can be assumed, with more plausible continuations showing a smaller N400 amplitude (Kutas & Federmeier, 2000). Conducting plausibility judgements of experimental items is recommended to ensure that no item-specific confounds affect the expected effect.

Additionally, Kutas and Federmeier (2011) also report that the N400 can be linked to long-term memory access as well as having a connection to the working memory. Intracranial recordings indicate that some brain regions that are activated in relation to the N400 are involved in long-term memory processes. Therefore, it seems that the amplitude is modulated not only by context but also by how easily information is retrieved from long-term memory (Kutas & Federmeier, 2000). Furthermore, the N400 has been observed not only in response to reading and listening to speech but also in reaction to line drawings (Federmeier & Kutas, 2001), photographs (McPherson & Holcomb, 1999), faces (Barrett & Rugg, 1989), environmental sounds (van Petten & Rheinfelder, 1995), and even odours (Sarfarazi et al., 1999).

Overall, changes in the N400 amplitude can be observed when the semantic context is altered in an unexpected manner or (semantic) memory needs to be engaged for retrieval. Both indicates a close link of the N400 to general semantic processing (van Petten & Luka, 2006). Thus, the study of the N400 might help to investigate how the brain processes semantic linguistic information, particularly during language comprehension tasks. This study aims to further address this question by proposing a framework for an experimental EEG study regarding semantic interference from sentence-external words. In the next section, a short overview about the concept of interference will be presented.

3. Interference

Effective language comprehension depends on the ability to retrieve non-adjacent constituents to form a reasonable semantic interpretation. This means that sentence processing can also be affected by certain factors. The two main influences on sentence processing are prediction and similarity-based interference. Prediction facilitates the processing of incoming information (Federmeier & Kutas, 1999), while interference hinders processing by interfering with memory retrieval (Jäger et al., 2017; Van Dyke & McElree, 2006). Although both processes involve memory representations, they operate through different mechanisms. Interference is usually linked to bottom-up processes which require the retrieval of previously encountered information. In contrast, prediction uses top-down processes by activating upcoming information based on anticipatory cues. Despite these differences, both processes use memory representations and can potentially interact (Husain et al., 2014).

For both, prediction and interference, memory processes play an important role, especially when establishing syntactic and semantic relationships between words. To comprehend sentence meaning, language users must derive meanings from representations that are no longer actively present in the mind or not the current focus of attention. The interconnectedness between language processing and memory systems allows language users to extract meaning from the ongoing linguistic input even if the grammatical dependencies do not appear in a linear order. Furthermore, it can be assumed that language parsing uses the same memory mechanisms as other cognitive tasks. Therefore, language processing should also depend on general memory limitations (Lewis et al., 2006; McElree, 2000).

The memory limitations that were observed in most complex psycholinguistic tasks have been generally attributed to working memory. The working memory is a cognitive system that can retain several items such as words, numbers or pictures simultaneously while also being able to allow the modification of these items. It has a limited capacity which explains the decline in performance when memory demands increases (P. A. Carpenter et al., 1990; Oberauer et al., 2000). Memory demand refers to the number of independent items that can be held in working memory simultaneously while still being accessible for

processing (Oberauer et al., 2000). The capacity of working memory is relatively consistent across a variety of tasks and has been attributed to limited resources (Oberauer et al., 2000), mutual interference from items held in working memory (Nairne, 1990) and limits of focus of attention (Cowan, 2001).

This underlines that successfully retrieving a dependent word relies on memory factors that can be influenced by processes such as interference or prediction. Despite the potential interaction between these processes, this thesis will mainly focus on interference and its effects on sentence processing. Interference processes occur when grammatical dependencies between words or phrases are being established but this process is disrupted by intervening words or phrases that interfere with those dependencies. This interference from other potentially relevant words can weaken the established link between two dependent words (Nairne, 2002a, 2002b). One form of interference is similarity-based interference which will be defined in the following subsection.

3.1 Similarity-based interference

The term similarity-based interference includes memory mechanisms that can lead to forgetting (Anderson & Neely, 1996; Nairne, 2002a) and assumes that language processing relies on memory mechanisms that are general. It suggests that forgetting does not occur because of the decay of an activation of a target item but because of the inability to retrieve a specific item from memory when it resembles other items stored there (Brown, 1958). This resemblance results in interference between similar items. Other items in a sentence (so-called distractors) can restrict the original item (called target) retrieval. The distractors have similar syntactic or semantic features to the target item and the matching features create a cue overload in memory. This overload inhibits the access to the target in memory and can often be observed in increased processing times. Interference seems to have the biggest effect on retrieval rather than on memory encoding or storage. The interference induces a cue-overload, where the cues that are needed to recover an item become associated with many different items in memory, making it unlikely to reliably choose a single corresponding target item (Jäger et al., 2017; Martin et al., 2012, 2014).

Two theories regarding the type of similarity-based interference have been developed. They both assume the same underlying retrieval mechanisms and assume that the shared features of the target item and the distractor item are responsible for the increased processing time (Jäger et al., 2015; Martin, 2018). However, the theories differ in how they define the retrieval cues. Whereas for the retrieval interference (also referred to as cue-based retrieval) only shared features that have been cued at the verb are defined as retrieval cues, in encoding interference all shared features between items can influence the interference between them. Although, both types of interference are distinct concepts, it can be challenging to differentiate between them since both impact the retrieval in a negative way (Villata et al., 2018). The following sections provide a brief overview of both theories.

3.1.2 Cue-based retrieval interference

Cue-based retrieval interference is a model of similarity-based interference that assumes that the integration of new words into the context relies mostly on memory retrieval. Memory retrieval is based on the assumption that each encountered word is encoded into memory as a set of features (such as animacy, gender or case). During retrieval, the cues provided at the retrieval site (usually the grammatical head) are used to identify the features that match with the features of words stored in memory (Jäger et al., 2017; Lewis et al., 2006; Van Dyke, 2007). It appears to be quite common that the retrieval site is the matrix verb (also called main verb) and thus retrieval cues are mostly triggered at the verb (Van Dyke, 2011; Van Dyke & McElree, 2006). But it is not only the current encountered word which is present in memory but also other words or grammatical constituents that have been activated through processes triggered by the recently encountered word. This pre-activation facilitates the integration of a word into its context and its processing (Jäger et al., 2017; Lewis et al., 2006; McElree, 2000). The retrieval of preactivated words from memory does not necessarily follow the linear order of the sentence. Instead, the memory system can access words based on their activation level and relevance to the demands of the current processing and not only by their immediate position in the sentence. This allows for the resolution of non-adjacent dependencies, where words or phrases that are not directly next to each other can still be linked correctly. This is important for more efficient sentence comprehension compared to simple word-to-word processing. Cue-based retrieval interference can be expected if multiple words in memory have

overlapping features regarding the retrieval cues that are expected by the retrieval site. In those cases, the retrieval of the correct word can be disrupted or slowed (Lewis et al., 2006; Van Dyke, 2011).

The main prediction of cue-based retrieval interference is that the likelihood of a successful retrieval of the target word depends on the degree of similarity between the target and its competing representations. This means that the more similar the distractors are to the target regarding the retrieval cues, the higher the probability of an incorrect retrieval and the longer the processing times are. This phenomenon of multiple items being activated simultaneously in memory is called cue-overload and is considered a main reason for retrieval failure (Engelmann et al., 2019). Challenges that are encountered during memory retrieval in specific regions of sentences, are often explained by retrieval interference.

Evidence for (cue-based) retrieval interference

Several studies provide evidence for retrieval interference. Van Dyke and McElree (2006) used a memory load paradigm with a self-paced reading task to investigate retrieval interference. In their experiment, the semantic retrieval cues at the matrix verb were manipulated to correspond to either only one specific noun in the sentence or also multiple nouns from the memory task. Their results showed increased reading times and reduced accuracy for the comprehension questions in the cases where the retrieval cues of the verb did not identify only one noun. These results suggest the possibility of semantic interference occurring during the retrieval stage. Another study providing evidence for retrieval interference focuses on the comprehension of object relative clauses in children. Belletti et al. (2012) showed that gender can function as a retrieval cue at the verb. Their result indicate that Hebrew-speaking children display higher comprehension accuracy for object relatives when the subject and the object differed in gender. The gender mismatch facilitates the disambiguation between object and subject since Hebrew verbs are gender-inflected. In contrast, in Italian, this effect was absent because verbs do not possess gender inflection. This provides additional support for retrieval interference.

A study by Van Dyke (2007) provides more evidence for cue-based retrieval interference through an eye tracking study. The study demonstrated that semantic interference from nouns structurally positioned between a verb and its target might occur even when the

syntactic position of the noun makes it an implausible target for the verb. The results showed longer regression path times at the end of sentences and reduced comprehension accuracy on the questions even when the distractor was an implausible target syntactically. Similar effects were observed when the intervening noun shared a syntactic role with the target and therefore making it a plausible target for the verb. For these effects to appear in both conditions, even when there are syntactic constraints that should exclude the intervening noun from being a target for the verb, it is necessary to override grammatical rules to arrive at a logical interpretation of the sentence. This suggests a retrieval interference effect because the cues of the verb needed to be satisfied and to achieve this, nouns that match the missing cues and are more recent in memory may be interpreted as the target word, even if that requires a syntactic violation.

But as Villata et al. (2018) pointed out, longer reading times or higher comprehension accuracy alone cannot be interpreted as evidence for retrieval interference just because they occur during the retrieval stage. The process of encoding interference might also impair retrieval and lead to reduced comprehension accuracy or incorrect retrievals. This study could provide a framework for investigating retrieval interference in relation to memory load, particularly through the use of EEG which might help in distinguishing between retrieval and encoding interference. The process of encoding interference will be explained after outlining two different approaches to cue-based retrieval interference.

3.1.2 Encoding interference

This theory of similarity-based interference assumes that the interference arises from competition between items that have similar features. It claims that interference between target items and distractors can occur regardless of whether the shared feature acts as a retrieval cue. It assumes that interference between words mainly arises during the encoding stage of the memory process, e.g. when encoding an item into memory previously encoded items with similar features might impede the encoding process (Fedorenko et al., 2006; Jäger et al., 2015). In Nairne (1990) this is explained as a process of feature-overwriting meaning that if two items share a feature, they are competing for this feature. If one item is proportionally less represented than the other one, it loses the memory representation of that specific feature. This means that the memory representation is less distinct for the losing item and thus, the retrieval probability is

lower. It is important to notice that even if the interference between those items happens at the encoding stage in memory, it can be seen at the slowdown of retrieval of that item at a later stage. This is due to the lower activation levels of the losing item in memory which can be seen in slower processing times (Nairne, 1990). It is possible to transform the degree of similarity between distractor item and the target item into a function that is able to describe the complexity of the encoding process. This function assumes that the higher the number of shared features between items that have to be remembered is, the more complex the process of encoding items into memory becomes (Fedorenko et al., 2006; Jäger et al., 2015). To summarise, the term encoding interference refers to any interference due to feature overlap where the overlapping features are not relevant for retrieval in the given context (Smith et al., 2021).

Evidence for Encoding Interference

Several studies provide evidence for encoding interference. Gordon et al. (2001) used a self-paced reading paradigm to investigate whether definite descriptions and pronouns influence the processing of object- vs. subject-relative clauses. Two types of sentences were tested that either contained two definite descriptions as noun phrases (conditions a and b) or one definite phrase and one pronoun as a noun phrase (conditions c and d). Apart from exchanging one definite noun phrase for a pronoun, the sentence structure in conditions a and c as well as b and d were identical. The results showed the well-known difficulty in processing object-relative clauses compared to subject-relative clauses. However, the difference was even more pronounced in the sentences that contained the definite noun and pronoun combination. Since pronouns and definite descriptions do not differ regarding their cues for the verb or in their syntactic relation to the verb, the processing difference between these two types of sentences cannot be explained by cue-based retrieval (Gordon et al., 2001; Gordon et al., 2004).

Similar results were also reported by Gordon et al. (2002) and Fedorenko et al. (2006) although the effects were detected at the retrieval region. Hofmeister and Vasishth (2014) also conducted a self-paced reading study in which they observed faster reading times at the verb in sentences with semantically and syntactically complex objects compared to simpler ones. Similarly to above, this complexity is not a retrieval cue and thus provides evidence for encoding interference.

However, a self-paced reading study by Jäger et al. (2015) trying to differentiate between encoding and cue-based retrieval interference in the processing of reflexive dependencies observed different results. In this study, participants were presented with sentences in which the subject of the matrix clause acted as the antecedent of the German third person reflexive “sich”. The subject was a noun phrase that could be either male or female and was followed by a relative clause that contained the distractor noun phrase. The distractor was either of the same gender as the subject or a different gender. The study found no online (real-time) evidence for encoding interference. Some offline evidence in the accuracy rates for comprehension questions in one of the three experiments could be interpreted as slight evidence for encoding interference. But interestingly, more regressions were observed during the gender mismatch conditions compared to gender match conditions. This could be a strong indicator for false retrieval of the interfering element and provides evidence for cue-based retrieval interference. Similarly, Schoknecht et al. (2023) found evidence for semantic interference which included both, encoding and retrieval interference, depending on the method used. In the self-paced reading paradigm, encoding interference was observed, while the EEG method seemed to point towards retrieval interference. The interference observed in the EEG was interpreted as retrieval interference since the effect appeared only at the critical word and not at preceding words which would be typical for encoding interference. Schoknecht et al. (2023) suggested that the difference in presentation rates between the self-paced reading and the fixed presentation times in the EEG study might have hidden possible signs of encoding interference in the EEG results. Hence, it seems that while cue-based retrieval and encoding interference are distinct theoretical concepts, it can be challenging to separate them empirically.

The current study aims to provide a framework for possibly examining proactive retrieval interference based on the influence of sentence-external memory items. Previous studies have shown that linguistic items which are not structurally integrated in a sentence still seem to be able to influence the parsing and prediction of a sentence and will be shortly presented in the next section (Fedorenko et al., 2006; Gordon et al., 2002; Van Dyke & McElree, 2006).

3.2 Retroactive versus Proactive Approach

Cue-based retrieval interference can be further divided into two types: the proactive and retroactive approach. The approaches differ in their assumptions about how the memory process is disrupted and the ability to recall information is disturbed (Crowder, 2014; Van Dyke, 2011).

The retroactive approach assumes that the interference between a target and its distractors is based on processes that happen between the process of encoding of the target word and its retrieval (Van Dyke, 2011). This form of cue-based retrieval interference typically occurs when the distractor is structurally positioned between the target word and its retrieval point during processing. This interference can negatively affect the memory and result in reduced encoding, storage or retrieval of the target word (Van Dyke, 2003; Van Dyke & McElree, 2006).

The proactive approach suggests that interference comes from processes that occur before the target word is initially encoded (Öztekin & McElree, 2007; Van Dyke, 2011). Specifically, interference can also occur when the distractors precede the target and the retrieval point of the target. For instance, if a semantically or syntactically similar word appears before the target has been encountered, this distractor word and its dependencies need to be processed before the target word can be processed. This prior processing of the distractor can still affect the encoding, retrieval or storage of the target word (Van Dyke, 2011; Van Dyke & McElree, 2006).

Van Dyke (2011) and Jäger et al. (2017) draw the conclusion that retroactive interference impacts dependency formation more significantly than proactive interference which suggests that interference effects are more noticeable when the distractor is located between co-dependent elements, such as subject and verb. Furthermore, Martin and McElree (2009) demonstrated that retroactive interference has a stronger impact on processing compared to proactive interference.

4. Previous studies

Several previous studies have investigated the influence of sentence-external factors on semantic processing and prediction, as well as the role of interference in relation to memory load.

Gordon et al. (2002) investigated the role of interference in a self-paced reading paradigm with memory load to allow for a direct manipulation of the contents of memory during comprehension. The participants were presented with a memory load prior to reading the sentence, and they were asked to remember that word list while reading. The size of the load remained consistent and the manipulation involved the similarity between items in the memory list and linguistic items in the sentence. For example, in the matched memory load condition, the memory list contained items similar to those used in the experimental sentence. In the mismatched memory load condition, the items in the memory list did not match the linguistic items in the sentence. The results showed that matched conditions had higher error rates and longer reading times compared to mismatched conditions which indicates that comprehension was more difficult when the items in memory matched the items in the sentence. However, this study did not provide insight into the underlying mechanism of the interference effect. It remains unclear whether the effects occurred during the encoding or the retrieval of the item. An extension to the Gordon et al. (2002) study was done by Fedorenko et al. (2006) by examining the impact of high vs. low memory load (one-word vs. three-word lists) on subject- and object-relative clause constructions. The results showed no effect of load for the first noun phrase, but slower reading times were observed for object-relatives under the high load condition. This suggests that similarity-based interference seems to play an important role in comprehension difficulty, especially when the memory load is high. However, similar to previous studies, it remains unclear whether this interference occurs during encoding or retrieval processes.

Another study using the self-paced reading paradigm has been done by Van Dyke and McElree (2006). They investigated cue-based retrieval interference from semantically similar and sentence-external distractors while the retrieval cues at the verb were manipulated. A cleft construction was presented to the participants and to correctly process this construction, it was necessary to retrieve the target from memory to integrate

the verb of the matrix sentence with its object. Two different conditions existed for this experiment. In the load condition, the subjects were presented with three distractor nouns that they had to memorise before reading the sentence. The load condition was further divided into high and low interference. High interference refers to the distractors matching the cues of the matrix verb and therefore being plausible objects. In the low interference condition the memory items did not match the matrix verb and were implausible objects. To exclude possible confounds, the control condition contained the same sentences but without any memory load. The reading times at the critical verb were measured. According to the cue-based retrieval account an interference effect should be visible because of a cue overload when the memory items match with the verb. The results showed longer reading times in the high interference conditions with memory load which confirms the prediction of the retrieval interference account. Hence, Van Dyke and McElree (2006) used the interaction between the memory load and the interference condition as evidence for a processing difficulty that is based on proactive retrieval interference from sentence-external distractors. The authors assume that the interference originates at retrieval since the semantic retrieval cues used during dependency formation can become overloaded. However, as mentioned in Mertzen et al. (2024), the study by Van Dyke and McElree (2006) showed statistically non-significant results, especially for the interaction between memory load and interference. Even in a follow-up study (Van Dyke et al., 2014) that used linear mixed models for the analysis of the interference effects no significant interaction could be observed.

But Campanelli et al. (2018) conducted a study with a design similar to that of Van Dyke and McElree (2006), investigating the interaction between interference and expectation using a self-paced reading paradigm. The results differed from Van Dyke and McElree (2006) insofar that no interference effects were observed in the critical region (at the verb), instead the effects emerged in the following spillover regions. Generally, sentences without a memory load were read more slowly than those with a memory load. Additionally, the response times to the comprehension questions showed that answering questions about memory-load sentences took longer compared to no-load sentences. These findings further support the memory interference effects reported by Van Dyke and McElree (2006).

Mertzen et al. (2024) also conducted a study which was based on the design of Van Dyke and McElree (2006) to investigate the interference from sentence-external factors in semantic processing. Their procedure involved a self-paced reading paradigm with additional eye tracking measurements. Here as well, the sentences were either preceded by the presentation of three memory nouns or by none. The main distinction from the Van Dyke and McElree (2006) study lies in the examined languages which are German, English and Russian as well as in the sentence structure. While Van Dyke and McElree (2006) tested object-cleft sentences, Mertzen et al. (2024) focused on non-clefted structures with two embedded relative clauses. The reasoning behind the usage of non-clefted sentences was to avoid linguistic focus on the target item which could have been a confound in Van Dyke and McElree (2006) clefted object sentences. Focus can impact memory representations because it assigns an emphasis or prominence to certain sentence constituents by the way of syntactic structure (Chomsky, 1996). According to Ward and Sturt (2007), items in focus possess more distinct memory representations, and thus could potentially reduce the magnitude of interference effects. The use of non-clefted stimuli in Mertzen et al.'s (2024) study might amplify the effect size and increase statistical power.

All sentences in the study included two embedded relative clauses, with the outer clause being an object-relative and the most embedded one being a subject-relative. Another distinction between Van Dyke & McElree (2006) and Mertzen et al. (2024) is the structure of the questions. Mertzen et al. (2024) wanted to test interference at different processing depths and asked either simple or complex comprehension questions to manipulate the processing depth. The results for memory recall indicated lower accuracy than in Van Dyke & McElree's (2006) study, possibly due to participants reading the memory items silently as opposed to reading aloud in Van Dyke & McElree's (2006) study. Conditions with memory load were generally being read faster than the no-load conditions. However, out of the three languages tested, German and Russian did not show an interaction effect between interference and memory load and English showed only very weak evidence for an interaction effect in the low processing depth (simple questions) condition. The authors concluded that it seems that sentence-external distractors do not influence or only slightly influence proactive retrieval interference.

To summarise, in a replication attempt of the original Van Dyke and McElree (2006) study, using almost the same design, items, and fillers, Mertzen et al. (2024) found

slightly lower memory recall rates and almost no interference effect. When statistical analyses were conducted with only those participants that showed high recall accuracy, the results showed greater similarity to Van Dyke and McElree (2006) findings. But even among high recall participants there was no indication of interference effects when items were held in memory. However, it could be possible that the observed effect might be too subtle to impact processing significantly and requires larger sample sizes or more sensitive methods to detect an effect.

These studies showing mixed evidence could be attributed to several factors. Firstly, as mentioned by Rabe et al. (2024) similarity-based interference effects are usually subtle. This is also highlighted by Jäger et al. (2017) in a metanalysis in which the interference effects show minimal impact with a range of 2 to 28ms. Additionally, interference effects may also remain undetected due to small participant sample sizes or be greatly inflated by statistical errors such as type M or type S errors (Jäger et al., 2020; Vasishth & Gelman, 2021). Type S error is defined as "the probability that the sign of the effect is incorrect, given that the result is either statistically significant or non-significant" (Vasishth & Nicenboim, 2016, p. 358). Type M (magnitude) error can be described as an error which appears when the estimated effect size in a study is exaggerated due to statistical noise. This can be a result of low statistical power and while the observed effect might be statistically significant, its size can be inaccurately large compared to the true effect (Vasishth & Nicenboim, 2016). Those two errors occur because psycholinguistic studies typically use participants sample sizes that range from 40 to 60 subjects, which can result in low statistical power (Vasishth & Gelman, 2021). The interference effect reported by Van Dyke and McElree (2006) was based on a sample size of 56 subjects. Seeing an effect in such a small sample size could likely be a result of inflation due to type M errors (Schoknecht et al., 2023) and this could explain the missing effect in Mertzen et al. (2024) since in their study a larger sample size was used which could minimise the appearance of a type M error. While the previous studies mentioned here were investigating interference effects of sentence-external items, none of them used EEG as the experimental paradigm. In the next section, previous studies investigating cue-based retrieval interference with EEG will be shortly mentioned.

4.1 Similarity-based retrieval interference in EEG studies

There are only a few EEG studies that have focused on the effects of cue-based retrieval interference. Martin et al. (2012) investigated the influence of distractors on the processing of noun ellipses in subject-extracted relative clauses in Spanish. The study manipulated the gender of the distractor so it would either match or mismatch the target, e.g. a female target with a male distractor. The results showed an increased negativity at 400ms (N400) for grammatical sentences with mismatching distractors compared to matched distractors which suggests that the features of the mismatching distractor disrupt the target retrieval in grammatical sentences. In an extension study, Martin et al. (2014) examined object-extracted relative clauses in which the target is less prominent than in subject-extracted relative clauses. This changes the retrieval cues that are required at the retrieval point for the target. Here, the determiner either matched or mismatched the gender of the noun. The results showed a reversed pattern from Martin et al. (2012): for grammatical sentences, the matched condition (same gender) elicited greater N400 negativity than the mismatched condition. The authors interpreted this finding as evidence for cue-based retrieval interference with gender as a retrieval cue even when the syntactic roles of the target and distractor are different. It seems that both, syntactic structure and gender cues can influence target retrieval.

Lee and Garnsey (2015) examined temporally ambiguous but grammatical relative clause sentences where the target noun was singular and the distractor noun plural. The authors observed a frontal negativity at the verb when the cues of the distractor matched the target and therefore the distractor was highly interfering. This suggests that the subject-verb dependency formation seems to be based on the cue-based retrieval interference account. When the verb is encountered, the memory representations are searched and the distractor, whose cues fully match the cues of the verb, can interfere even when the distractor violates the number agreement (singular vs. plural). This can disrupt retrieval by either causing the distractor to be erroneously selected or by diverting attention away from the correct target. Both studies investigated cue-based retrieval interference and found supporting results but none examined the influence of sentence-external factors on interference. The following section will shortly present studies that again only focused on interference effects without considering sentence-external factors but those studies are focused on German which will also be the investigated language in the proposed study.

4.2 Similarity-based interference effects in German

Relatively few studies have investigated interference effects in German and even fewer have used the EEG paradigm. Nicenboim et al. (2018) investigated number interference effects on nouns in a self-paced reading task in German. The results show that reading times were longer in those conditions where target and distractor had the same number feature compared to conditions where there was a number mismatch between target and distractors. Their findings provide (limited) evidence for interference effects and suggest that the retrieval of the target is hindered when the distractors share the same number features. However, given the small magnitude of the observed effect it is likely that number agreement is not a highly significant cue for retrieval.

Schoknecht et al. (2023) is one of the few studies investigating syntactic and semantic interference in German using the EEG and self-paced reading paradigm. The results show clear evidence of semantic interference but no indication of syntactic interference. The study examined animacy as a retrieval cue with the high interference condition including both the target and distractor as animate nouns, whereas in the low interference condition the distractor was inanimate. The analysis focused on the critical region (the verb), the spillover region, and the pre-critical region. In the self-paced reading paradigm, semantic interference was observed across all three regions, with longer reading times in the high interference condition compared to the low interference condition. These findings differ from those of Mertzen et al. (2024) and Van Dyke (2007) since Schoknecht et al. (2023) observed no syntactic interference effect. But they observed a semantic effect in all three regions whereas in Mertzen et al. (2024) a semantic interference effect in German could only be found in the pre-critical region. Potential explanations suggested by the authors include methodological differences (self-paced reading versus eye-tracking), variation in sample sizes since Schoknecht et al. (2023) used a considerably larger sample, and differences in item structures. For the EEG part of the study, the results showed a clear N400 effect that could be interpreted as semantic interference. Specifically, the N400 amplitude was more negative in high interference conditions compared to low interference conditions which seems to indicate increased complexity of memory retrieval under high interference.

5. Current study

The mixed evidence for interference effects as well as the only partial evidence for cue-based retrieval interference that has been observed in the studies mentioned above motivated the design of the study that is proposed here. The objective of this thesis is to establish a framework for an EEG study that focuses on semantic interference within the context of memory load. Specifically, it investigates proactive semantic interference because of sentence-external distractors. This thesis aims to outline the required methodological structure to extend previous findings (e.g. Van Dyke and McElree (2006); Mertzen et al. (2024)) by using EEG measurements to capture online processing data. The use of the EEG method could offer further insights into the processing of semantic interference, memory mechanisms as well as help in differentiating between encoding and retrieval interference. This thesis provides a resource for future studies, including experimental items and fillers, a foundation for statistical ERP analysis, and a simulation of potential EEG data outcomes in a semantic interference paradigm involving sentence-external distractors. Additionally, it provides interpretations of potential results and discusses limitations when conducting EEG studies within this paradigm. Therefore, it may serve as a reference for future EEG research on the interaction of memory and semantic interference in German. To date to the best of my knowledge, no study has directly examined the impact of sentence-external factors on proactive interference in German using EEG methodology.

5.1 Motivation

This thesis aims to provide a framework for replicating the study conducted by Van Dyke and McElree (2006), which investigated semantic interference with the self-paced reading paradigm in English. The reported results suggest the presence of sentence-external semantic interference in sentence processing. Mertzen et al. (2024) replicated this study while also examining the phenomenon of sentence-external semantic interference in other languages by using self-paced reading and eye tracking. However, this study found no support for semantic interference from sentence-external factors. Another self-paced reading study by Van Dyke (2007) reported evidence for both syntactic and semantic

interference in sentence processing, which could be partly replicated in Mertzen et al. (2024) by describing evidence for syntactic interference. However, a follow-up study by Schoknecht et al. (2023) using ERP measurements found evidence for semantic interference but not syntactic interference, contradicting the findings of Mertzen et al. (2024) and partly supporting the findings of Van Dyke (2007). It appears that three self-paced reading and eye tracking studies report contrasting results, with two being (partially) supported by an EEG study. Given the conflicting evidence from these studies for the existence of semantic interference from sentence-external factors further research on this phenomenon is needed.

Another significant aspect is that, thus far, no study has investigated the possibility of sentence-external semantic interference by using ERP measurements. Tanner et al. (2014) highlighted that online methods such as EEG are important in identifying processing difficulties that could remain undetected with offline methods such as self-paced reading. It is possible that memory retrieval effects might not be observed in offline methods but could be detected in ERP measurements. One advantage of ERPs is their ability to show the varying sensitivity of the brain to specific linguistic manipulations with high temporal resolution that allows to track real-time retrieval during sentence comprehension (Gaskell, 2007). Additionally, ERPs provide insights into the temporal dynamics of sentence processing and its components. This might not only help in identifying the specific location in the sentence where effects occur but might also highlight processing difficulties marked by certain ERP components, such as the N400 (Tanner et al., 2014).

Furthermore, while much of the existing research on similarity-based interference has concentrated on English, this thesis focuses on German. As noted in Schoknecht et al. (2022) German is a morphologically rich language and explicitly marks features such as number and gender. This makes German particularly interesting to research in this context.

6. Methods

This thesis describes a simulation of the intended study since the process of collecting data for EEG studies is highly time-intensive and exceeds the timeframe available for a Master thesis. However, the proposed procedure, participant criteria, and methods are outlined here as preparation for the eventual implementation of the experiment.

6.1 Participants

If this study were to be conducted in a laboratory setting, it would be essential to define clear participant inclusion criteria. For an EEG study of this type, recruitment should focus on monolingual German speakers aged 18 to 40 years who are right-handed and do not have neurological impairments. Ideally, the participant sample would include an equal number of males and females with an average of 29 years. However, these demographic specifications should serve as a general guideline rather than strict requirements due to practical challenges in meeting them in experimental studies. Based on a power analysis[1] in the statistical software R (R Core Team, 2024), a sample size of 120 participants would provide 100% probability of detecting an effect at the upper bound and 80% for the mean value, both indicating a strong likelihood of detecting an effect. For comparison, power levels for sample sizes of 100, 80 and 60 participants could also be considered (see Table 4 in the section "Power analysis").

6.2 Materials

All materials and codes for this thesis can be accessed through the OSF repository at the following link: https://osf.io/3sugm/?view_only=fe57c3e42e7c439fb08bcab5b1c31983

6.2.1 Experimental items

The experiment includes 120 items distributed across four conditions. The items were constructed by partially repurposing materials from Mertzen et al. (2024) and the sentence structure is based on the cleft sentence structure in Van Dyke and McElree (2006). By the

[1] For a more comprehensive explanation refer to subsection "Power analysis" further below.

time this thesis is handed in, some of the experimental items are still under revision. The experiment uses a 2 × 2 design similar to Van Dyke and McElree (2006). Items are presented by using a Latin square design to ensure that each participant encounters each item only once, within a single experimental condition. There exist two main conditions: one with a memory load where participants are required to hold three inanimate nouns in memory prior to reading a sentence, and a control condition with no memory load. The main conditions were further divided into two sub conditions: high and low interference. In the high interference sub condition, the memory items semantically match the matrix verb, whereas in the low interference sub condition, there is no semantic match between the matrix verb and the memory items. To check for sentence comprehension, half of the experimental items in each session are followed by a comprehension question.

In Table 1 an example item can be seen. For the sentence comprehension task, the noun "Boot" needs to be retrieved at the non-adjacent matrix verb "ruderte/reparierte" to ensure correct sentence understanding. During this retrieval process the memorised memory items "Zaun, Uhr, Rad," act as sentence-external distractors that could potentially interfere. This interference was manipulated based on the degree of match between the distractor items and the matrix verb. There are also two conditions without a memory load to control for potential confounds.

Condition	Memory list	Sentence	Comprehension Question
Low interference, memory load	Zaun - Uhr - Rad *Fence - Clock - Wheel*	Es war das Boot, das der Mann, der am Meer lebte, gestern ruderte. *It was the boat that the man who lived at the sea rowed.*	Lebte der Mann am Meer?
High interference, memory load	Zaun - Uhr - Rad *Fence - Clock - Wheel*	Es war das Boot, das der Mann, der am Meer lebte, gestern reparierte. *It was the boat that the man who lived at the sea repaired.*	Lebte der Mann am Meer?
Low interference, no memory load	—	Es war das Boot, das der Mann, der am Meer lebte, gestern ruderte. *It was the boat that the man who lived at the sea rowed.*	Lebte der Mann am Meer?
High interference, no memory load	—	Es war das Boot, das der Mann, der am Meer lebte, gestern reparierte. *It was the boat that the man who lived at the sea repaired.*	Lebte der Mann am Meer?

Table 1: Example item for the experimental items.

Because it might be possible that the experimental items show a high variability in the acceptance rates, this should be assessed before conducting the study. This could be done

by using a platform that offers their service to a wide range of participants to ensure participants diversity. This cross-checking of the experimental items can minimise the risk of the item variability influencing the results and possibly masking interference effects.

6.2.2 Filler

The filler dataset consists also of 120 items, of which half include a comprehension question. Of these 60 comprehension questions, 30 should be answered in the affirmative and 30 should be answered in the negative. The filler items are taken from Mertzen et al. (2024) and two example fillers can be seen in Table 2. The filler dataset features 60 subordinate clause-main clause constructions in which each subordinate clause begins with a conjunction. Additionally, the filler dataset includes 60 main clause–main clause constructions. Half of these filler items are preceded by three memory words, while the remaining half is not.

Filler Type	Memory list	Sentence	Comprehension Question
Subordinate clause - Main clause	Stuhl - Baum - Kochtopf *Chair - Tree - Pot*	Dass der Junge den Ball warf, erfüllte die Eltern mit Stolz. *That the boy threw the ball filled the parents with pride.*	Warf der Junge den Ball? *Threw the boy the ball?*
Main clause - Main clause	Wolle - Kabel - Zahnbürste *Wool - Cable - Toothbrush*	Der Mann konnte nicht schlafen und machte einen langen Spaziergang. *The man couldn't sleep and took a long walk.*	Machte er einen Spaziergang? *Took he a long walk?*

Table 2: Example items for the filler items.

6.3 Procedure

The following procedure might offer guidance for future EEG studies on this topic. Each participant will encounter 240 sentences in total of which 120 are target items and 120 are filler sentences. The sentences are presented in a pseudo-randomised order. To avoid prolonged experimental sessions and therefore potential loss of participant focus, the experiment should be conducted over two separate sessions. The task should be divided into six blocks with each block containing 20 sentences and with brief breaks between blocks to reduce fatigue. Each recording session will last approximately two hours with a minimum interval of one week and a maximum interval of eight weeks between sessions.

At the beginning of each session, participants should complete a participant information sheet and sign a consent form. Following this, the EEG cap will be fitted, and the procedure will be explained. An initial recording of resting-state ERP activity will establish an individual baseline for each participant (Klimesch, 1999). Then, a few practice trials should be carried out to confirm that the participant understood the task instructions. Each session should be conducted in a soundproof booth with dimmable lighting to prevent distractions that might introduce noise into the ERP signal (Luck, 2014).

After ensuring the instructions were understood correctly, the experimental items will be presented. The studies by Schoknecht et al. (2023) and Van Dyke and McElree (2006) were used as a guidance for presentation times which might need to be adjusted. First, a fixation cross will appear at the centre of the screen for 500 ms. Following this, the three memory words (if applicable) will be shown simultaneously in the centre of the screen for 3 seconds, arranged on one line separated by five dashes. Following the method used by Van Dyke and McElree (2006), participants will read the memory items aloud and memorise them. The sentence then appears in a phrase-by-phrase format, with each phrase displayed for 300 ms plus an additional 20 ms per character, followed by a 400 ms interval between phrases. After the sentence, participants will respond to a comprehension question in writing if one is included for that item. Next, they will be prompted to recall and type the memory items in the correct order to ensure accurate recall of the words. There is no time limit for this recall section, and participants can skip words they do not remember. Each new trial begins immediately after the final memory word is entered. The EEG will be recorded continuously throughout the trials.

6.4 Region of interest

In the self-paced reading study by Van Dyke and McElree (2006) the interference effects were mainly visible in the critical and the post-critical (spill-over) region and a minor effect could be observed in the region that immediately follows the presentation of the memory distractor items. By contrast, in the self-paced reading study by Schoknecht et al. (2023) an interference effect could be observed after participants read the distractor and this effect continued through all following regions. The ERP data from Schoknecht

et al. (2023) showed no significant interference effect prior to the critical region (verb) and a significant main effect was observed at the critical region. Based on these findings, the region of interest in this experiment should focus on the processing of the verb within the relative clause. Thus, for the EEG data in this study it is recommended to focus the analysis on the ERPs that appear within the time window of the critical verb. Additionally, the analysis could benefit from examining the two preceding regions of the critical region to ensure no effects can be observed in those areas. Since the critical verb is at the end of the sentence, no spillover effects to the following words can be expected.

6.5 Hypotheses

Building on the assumptions from Van Dyke and McElree (2006) and Mertzen et al. (2024) it is assumed that an interaction between memory load and interference can be observed. It is expected that faster reading times reported by Van Dyke and McElree (2006) should correspond to more negative EEG amplitudes in this study. The expected results are summarised in Table 3 and suggest that in the memory load condition a significant difference in the N400 amplitude between the low and the high interference condition is expected. The N400 amplitude should be more negative in the high interference condition due to interference of the memory items as distractors. In contrast, in the conditions without memory load no significant differences between the low and high interference condition are expected to be observed.

Condition	**Reading times Van Dyke and McElree (2006)**	**Expected Amplitude**
Low interference, memory load	540 ms	Most positive amplitude
High interference, memory load	578 ms	Most negative amplitude
Low interference, no memory load	567 ms	Second negative amplitude
High interference, no memory load	564 ms	Second positive amplitude

Table 3: Expected N400 amplitudes in the EEG study based on the reading times in the critical region in Van Dyke and McElree (2006).

7. Data simulation

The data simulation and Bayesian analysis were carried out by using the statistical software R (R Core Team, 2024) with the lme4, brms and rstan packages (Bates et al., 2015; Bürkner, 2017; Stan Development Team, 2024). The simulation code is partly adapted from a code provided in a video by Vasishth (2020, June 14). The code used for the simulation, along with the simulated data points and further analyses is available at the OSF repository for this thesis.

In this thesis a simulated EEG dataset is used to investigate the effects of memory load, interference, and their interaction on ERP amplitude by focusing on the N400 effect in sentence processing. The simulations are based on a Latin square design with four conditions and include parameters that were derived from a prior EEG study by Schoknecht et al. (2023). Because Schoknecht et al. (2023) identified only the semantic interaction to be significant, this was the effect included in the simulation. For the generating of realistic EEG responses in simulated participants the original dataset from Schoknecht et al. (2023) was taken as a basis to fit parameters for a maximal linear mixed-effects model. The simulated datapoints represent ERP responses that could be observed in the critical region and are expected to appear within the typical time window of an N400 effect. The dataset was generated for 120 items and 120 participants although simulations with 60, 80 and 100 participants were also conducted to assess the effects in smaller sample sizes.

The simulated data was structured in a Latin square design with four conditions (a, b, c, d), where each condition represents a combination of memory load (high/low) and interference (high/low). The Latin square design ensures that each condition is evenly distributed across subjects and items which helps to control for potential confounding effects. The contrast coding for the conditions was defined as follows: In the memory condition, the coding of -0.5 was used for condition “b” since it represents the highly interfering memory load condition and therefore, the most negative N400 amplitude can be expected here. Condition “a” was coded as +0.5 since it represents the highly interfering no memory load condition. For the interference condition, again condition “b” represents the high interference memory load condition and is contrast coded with -0.5 since the most negative N400 amplitude is expected here. Condition “d” represents the

low interference memory load condition that is expected to have a more positive amplitude than condition “b” and therefore is coded as +0.5. In the interaction condition, condition “b” (high interference and memory load) is contrasted with condition “c” which represents low interference without memory load. Here as well, it can be expected that the N400 amplitude will be the most negative for condition “b”, hence its contrast coding as -0.5. The interaction term was included as a variable to investigate the combined effects of memory load and interference and is the main focus of this study.

Additionally, the simulated data includes EEG-specific details such as the region of interest (ROI) that was predefined as "centro-parietal". This region was selected because in actual experimental EEG studies the N400 effect can typically be measured at the centro-parietal electrodes (Kutas & Federmeier, 2000, 2011; Schoknecht et al., 2023).

The simulated data assumes that there exists variability in the ERP amplitude measurements and integrates this in the simulation by defining random slopes and intercepts for subjects and items. The random intercepts allow for differences in the baseline EEG amplitudes across participants and experimental items. The random slopes for memory, interference and their interaction allow for individual differences between participants, as well as item-specific differences. Those random effects for subjects and items were created by using multivariate normal distributions at the participant and item levels. The covariance matrices for these effects were constructed based on the standard deviations and correlations estimated from prior real EEG data from Schoknecht et al. (2023). The random effects were decided to be uncorrelated because this simplifies the model by assuming that no covariance between the intercepts and slopes for each factor exist. This helps to avoid overfitting the model and also to improve the efficiency of the computation. Since the correlation between the random effects is not central to the research question it was chosen to be ignored.

For each simulated observation an amplitude value that was based on the combination of fixed effects (memory, interference, interaction) and random effects was generated. Then Gaussian noise was added and standardised with a standard deviation that was estimated from the prior model fit of Schoknecht et al. (2023). This was done to simulate the natural variability and error that can be observed in real EEG data. The following formula describes how the simulated data was generated with the β coefficients representing the

fixed effect sizes, u_{subj} and w_{item} describing subject- and item-level random intercepts and $\epsilon_i \sim N(0, \sigma_e^2)$ describing the residual noise.

$$amplitude_i = \beta_0 + u_{subj} + w_{item} + (\beta_1 + u_{subj,memory} + w_{item,memory}) * memory_i + (\beta_2 + u_{subj,interference} + w_{item,interference}) * interference_i + (\beta_3 + u_{subj,int} + w_{item,int}) * int_i + \epsilon_i \sim N(0, \sigma_e^2)$$

To assess how robust the effect is across various effect sizes, 100 simulations for each range of estimates (lower, mean, and upper bound) were run. A linear mixed-effects model was fitted for each simulated dataset with the lmer function (Bates et al., 2015) to estimate the effects of memory, interference and their interaction on the amplitude of the ERPs. Each model was checked for potential convergence issues and the simulations that failed to converge were marked. The results of the analysis, such as the effect sizes and t-values for memory load, interference, and the interaction were saved and then used for further analysis of the data. The saving of the values ensures that the analysis is reproducible. Furthermore, the average amplitude for each range of estimate across conditions was calculated and then averaged across all simulations so a summary for each bound could be reported.

8. Data analysis

8.1 Power analysis

The power analysis was based on the t-value matrices that were extracted for memory, interference and their interaction as well as the corresponding effect size (lower, mean and upper bound) estimates from the simulations. Despite calculating power for all three effect sizes, the main focus of the power analysis was on the interaction between the memory load and semantic interference conditions. The interaction represents the main interest of this study and only those power values are reported here.

The statistical power was calculated as the proportion of absolute t-values from the simulated data that exceed the statistical significance threshold of greater than 2 ($|t| > 2$) with missing values (NA) being excluded from the calculation. The power analysis was based on the assumption of a two-sided t-test which uses a 95% confidence interval. This means that there exists a 2.5% probability for each tail of the distribution to incorrectly conclude that a significant interaction between memory and interference exists due to a Type I error. A Type I error occurs when the null hypothesis is rejected when it is actually true (Gelman & Hill, 2021).

Bound	60 participants	80 participants	100 participants	120 participants
Lower	4.30 %	15.38 %	5.43 %	7.05 %
Mean	51.55 %	69.51 %	74.16 %	80.00 %
Upper	97.70 %	98.88 %	100.0 %	100.0 %

Table 4: Power calculations for different sample sizes for all three bounds.

Table 4 shows the power estimates for detecting an interaction effect in the study for varying sample sizes (60, 80, 100, and 120 participants). The power values are rounded to two decimal places and include the lower bound, mean, and upper bound of the estimates. It is clearly visible that when the sample size increases, the mean power steadily increases as well and reaches the threshold of 80% at 120 participants. The threshold of 80% was chosen for the power analysis of the current study as it represents a accepted standard in research for balancing the risks of Type I and Type II errors. A Type II error occurs when a statistical test fails to detect a true effect which then leads to the incorrect acceptance of the null hypothesis (Gelman & Hill, 2021). An 80% power

level indicates that the study has a high probability of detecting a true effect if one exists while still ensuring that the study remains manageable regarding sample size and required resources (Cohen, 2013).The upper bound seems to be consistently around 100% which indicates a high probability of detecting an interaction effect with larger sample sizes.

The power analysis helps in evaluating the likelihood of detecting statistically significant effects for the interaction condition for different effect sizes (lower, mean and upper bound). This is useful for assessing the sensitivity of the experimental design and the reliability of the statistical tests under different effect size scenarios.

8.2 Choice of priors

In Bayesian modelling priors express beliefs about the parameters in the model before the data has been observed (Nicenboim et al., 2024). The selection of priors for the Bayesian model was based on prior EEG studies focusing on semantic interference. As noted in Gelman et al. (2013), the choice of priors is very subjective and can allow researchers to influence the model outcomes in a way that could correspond to the anticipated or preferred results. Studies such as those by Holtmann et al. (2016) have shown that poorly chosen priors can bias the model results, especially when the prior beliefs are inconsistent with the observed data. In such cases, the data must be especially strong to counteract the influence of these priors. On the other hand, other studies (Morris et al., 2015) have demonstrated that well-chosen priors can support and strengthen the data. Furthermore, priors can combine insights from previous studies with new data and update beliefs on a research topic (Gelman et al., 2013).

Bayesian priors can be grouped into three main categories. One category is non-informative priors that are intentionally broad and contribute little or no additional information to the experimentally observed data (posterior distribution) beyond what is already observed in the data. In these cases, the likelihood function strongly influences the results of the model. Non-informative priors are often used when prior knowledge about a parameter is limited or when subjectivity in the analysis should be reduced. They are very valuable for exploratory analyses (Gelman et al., 2013; Reis et al., 2023).

Another category is informative priors which specify a range for a parameter that is based on prior beliefs about the expected characteristics of the data. These beliefs should be based on previous empirical research such as meta-analyses or large-scale studies which provide a strong justification for the assumptions about the properties of the parameter. However, if these informative priors are misjudged there is the possibility of introducing bias into the model which could lead to inaccurate results and skewed estimates (Gelman et al., 2013; Reis et al., 2023).

A third approach is to use weakly informative priors. This approach incorporates the benefits of non-informative and informative priors since it includes mild constraints that based on prior research findings (Reis et al., 2023). Usually, this type of prior starts with a non-informative prior and then slowly narrows the range down to include values that are plausible for the studied phenomenon. For example, it would be useful to avoid extreme values for the variance and focus the main part of the probability towards the centre of the distribution (Gelman et al., 2013). Weakly informative priors are recommended when the available data is insufficient to justify strongly informative priors but is too comprehensive for non-informative priors (Reis et al., 2023).

To determine whether sentence-external items can affect semantic interference during sentence processing, it is necessary to compute the Bayes factor (see section Bayes factor for further details). As discussed by Vasishth et al. (2023) it is important to conduct a sensitivity analysis when calculating the Bayes factor. This approach ensures a robust assessment of the influence of different prior assumptions and model specifications on the interference effect. It is also used to verify that the selected priors are appropriate. A sensitivity analysis is conducted by running a model with priors that vary in their degree of informativeness. Afterwards, it is possible to examine how changes in prior distributions or model assumptions could potentially affect posterior estimates and how robust the model is. It is especially useful when the prior knowledge is limited or subjective (Reis et al., 2023). In a typical sensitivity analysis, multiple reasonable priors are tested to observe their impact on the posterior distribution. If the posterior results change significantly based on the chosen prior, it could suggest that the conclusions that can be reliably drawn from the model are limited (Schützler & Schlüter, 2022).

In this thesis, a sensitivity analysis was conducted using the simulated data. The analysis included four different prior specifications with varying levels of informativeness.

Following an approach suggested in Vasishth et al. (2023) a review of existing literature helped in identifying suitable priors for the analysis. Given the close similarity between the effects investigated in Schoknecht et al. (2023) and the current study, identical priors were used for the sensitivity analysis. Only the prior for the interaction term was varied, the priors for all other effects remained unchanged. Previous literature (Jäger et al., 2017; Rabe et al., 2024; Schoknecht et al., 2023) regarding interference effects showed that similarity-based interference effects are usually subtle and only show changes of a small magnitude, for which it is recommended in Vasishth et al. (2023) to use relatively informative priors.

In this sensitivity analysis, the priors for all parameters except the interaction slope were set at plausible ranges while avoiding overly restrictive assumptions. In Table 5 the priors for all parameters can be seen and again the R (R Core Team, 2024) code can be found at the OSF repository. The prior for the intercept was set to range between -10 to +10 microvolt (µV) to ensure a reasonable baseline amplitude. The slopes fore memory load and interference were assigned priors ranging from -4 to +4 µV with the mean of the distribution centred around 0. Here, a mean centred around 0 means that there exists no initial bias towards a negative or positive effect (Schad et al., 2021). For the interaction effect the slope priors were varied to conduct a sensitivity analysis. The most informative prior assumed a very narrow distribution for the coefficients with a standard deviation of 0.1 µV. An informative (standard deviation of 0.5 µV) and moderately informative (standard deviation of 1 µV) prior were added to relax this constraint and the broad prior was the least restrictive with a standard deviation of 2 µV. In all cases the mean of the prior distribution was centred around 0. For each level of prior informativeness a null prior model was defined that excluded the interaction term to act as a baseline for hypothesis testing (for further details see section Bayes factor). The priors for the residual standard deviation (σ) ranged from -5 to + 5 µV with a mean value of 10 µV. This weakly informative prior allows the posterior to adapt to the simulated data but still excludes extreme residual standard deviation. The prior for the standard deviations of the random effects ranged from -2 to +2 µV, with a mean value of 0. This prior assumes that there is only moderate variability across subjects and items. Next, a Bayesian model was fitted.

Parameter	**Prior**	**Assumed Range (in µV)**
Intercept	normal(0, 5)	(-10, 10)
Slope		
Memory	normal(0, 2)	(-4, 4)
Interference	normal(0, 2)	(-4, 4)
Interaction		
Highly informative	normal(0, 0.1)	(-0.2, 0.2)
Informative	normal(0, 0.5)	(-1, 1)
Moderately informative	normal(0, 1)	(-2, 2)
Broad	normal(0, 2)	(-4, 4)
Sigma	normal(10, 5)	(0, 20)
Standard Deviation	normal(0, 2)	(-4, 4)

Table 4: The prior specifications and the corresponding microvolt ranges for each parameter used in the computation of the Bayes model.

8.3 Bayesian analysis

All R codes (R Core Team, 2024) used in this section can be found in the OSF repository for this thesis.

First, the simulated datasets for 100 and 120 participants were imported. Both datasets were used for the analysis since in the power analysis it has been shown that the upper bound for 100 participants seems to reflect 100% probability to find an effect which could potentially minimise the effort of recruiting participants. For both datasets, the mean and standard deviation of the amplitudes were calculated, and the distribution was visually inspected to determine whether it approximated a normal distribution. Since the inspection confirmed that the data followed a normal distribution the default Gaussian family in R (R Core Team, 2024) was appropriate for the analysis.

A Bayesian mixed-effects model was fitted to investigate the interaction effect of memory load and interference on ERP amplitudes. This model used the brms package in R (Bürkner, 2017) which provides a user-friendly interface to allow for the implementation of Bayesian models with the programming language STAN (B. Carpenter et al., 2017). The fitted model included fixed effects for memory load, semantic interference, and their interaction, with random intercepts and slopes at the subject and item levels to account for variability at subject and item levels. The model used the Markov Chain Monte Carlo (MCMC) sampling (Gamerman & Lopes, 2006) which is a standard sampling method in statistical modelling with STAN.

The following paragraph shortly explains how MCMC sampling, including specific algorithms, functions for the better understanding of readers that might not be highly proficient in statistical analysis with STAN in Bayesian modelling.

MCMC sampling generates a sequence of samples in which each sample depends on the previous one while still making sure that the samples will converge to the true posterior distribution over time. In this model, the Hamiltonian Monte Carlo (HMC) method is used since it is effective for computing complex and high-dimensional distributions that have multiple fixed effects, interaction terms and random effects which is the case in the current model. To improve the efficiency of the computation, the No-U-Turn Sampler (NUTS) is used because this makes it possible to avoid the manual tuning of the number of steps that need to be taken by the HMC sampler (Hoffman & Gelman, 2014). Because

normally in the HMC method, it is necessary to manually specify the number of steps that the sampler takes during each iteration. In this case, steps refer to leapfrog integration that simulates the movement of a particle through the landscape of the posterior distribution and each step involves the update of the position and the momentum of the particle (Betancourt, 2017). If the number of steps is too small, the posterior distribution might not be explored thoroughly enough. But if too many steps are taken, the computational process might become unnecessarily time-consuming since each step requires computational effort. Moreover, the sampler may overshoot and not sample adequately from posterior regions that have a high density since the trajectory of the steps continues into less relevant areas of the parameter space. The NUT sampler makes the usage of HMC easier since it automatically decides, based on the distribution and geometry of the posterior, how many steps are necessary. It monitors the trajectory of the particle and once it starts taking a U-turn in the parameter space it stops automatically and prevents inefficient sampling (Hoffman & Gelman, 2014).

Furthermore, four sets of priors for the interaction effect were included in the analysis (refer to section priors for further specification) with varying degrees of informativeness to conduct a sensitivity analysis. The simulated data points were used for fitting the Bayesian model and it was run with 20,000 iterations with the first 2,000 iterations being discarded as warm-up so a stationary posterior distribution can be reached (Walsh, 2004). The remaining 18,000 iterations are used for the model sampling. The number of iterations was set high on purpose even despite the increased computation time because Nicenboim et al. (2024) highlighted that the stability of the results of the Bayes factor depends on the amount of posterior samples obtained. If this number is too low, it can lead to instability and inconsistency in the Bayes factor estimates, especially when using bridge sampling (for an explanation of this method, see below). Four computational cores were used for parallel processing to improve the efficiency of computation (TEMoore, (n.d.)) and the priors were sampled alongside the posterior distribution to make prior predictive checks possible. Model fitting was repeated for all prior levels and for the null models. The fitted Bayesian mixed-effects model was saved so it can be used for future analysis without re-running the fitting process.

9. Results

9.1 Bayes factor

The Bayes factor provides a measurement about which model is more strongly supported by the data. A value greater than one indicates stronger support for the full model while a value less than one indicates stronger support for the null model, a value close to one suggest that both models are equally likely. A Bayes factor greater than 10 indicates substantial evidence for the full model compared to the null model (Jeffreys, 1939, 2003; Nicenboim et al., 2024).

The Bayes factors for the models fitted in this thesis are computed by using two different methods: hypothesis testing and bridge sampling. Both methods assess the strength of evidence in favour of the null hypothesis (absence of an interaction effect) relative to the alternative hypothesis (the interaction term is present and significant). For both methods the Bayes factor was computed across multiple sets of priors to perform a sensitivity analysis. Initially, the hypothesis method was used to compute the Bayes factor for a sample size of 100 participants. However, as can be seen in Table 6 the results were anomalous, returning extremely high or negative values.

Prior	**Hypothesis testing**	**Bridge sampling**
Highly informative	-7081826139020562289282 62	537.91
Informative	14673050699302988	669463.76
Moderately informative	6044990183935335	587301.17
Broad	825.50	337713.42

Table 5: Bayes factors for the different prior specifications for a sample size of 100 participants for the method of hypothesis testing and bridge sampling.

To ensure this was not due to methodological issues, the bridge sampling method was also applied but despite using a different approach irregular results were observed again. To test whether those anomalous results might have been caused by the sample size, both methods were run with a sample size of 120 participants and both produced consistent and reasonable Bayes factors with that sample size. This suggests that the observed

anomalies might be based on computational issues due to an insufficient number of iterations or possibly some numerical instability in the posterior distribution that can be caused by the smaller sample size. It seems to be the sample size that influences those results and this again highlights the importance of sample size for a reliable Bayes factor computation. The conclusion that can be drawn here, after computing the Bayes factors for both sample sizes, is the recommendation of including a minimum of 120 participants because this ensures reliable results that can be correctly interpreted when using the appropriate statistical measures. In the next section, both computational methods as well as their results for 120 participants will be reported.

9.1.1 Hypothesis testing method

The calculation of the Bayes factor using the hypothesis testing method is based on the Savage-Dickey density ratio method (Wagenmakers et al., 2010) which calculates the Bayes factor as the ratio of the prior density compared to the ratio of the posterior density at the null hypothesis value instead of using the marginal likelihoods (Wagenmakers et al., 2010). This method is used to compare the null hypothesis and alternative hypothesis within the same model and therefore, the Bayes factor provides a straightforward interpretation for how much evidence the data offers to support or contradict the null hypothesis (Nicenboim et al., 2024). For example, a Bayes factor of 2 means that the data provides twice as much evidence in favour of the alternative hypothesis compared to the null hypothesis. In this analysis the null hypothesis was set at 0, assuming no interaction effect between memory load and semantic interference compared to the control conditions. The Savage-Dickey method was implemented in the analysis by applying the hypothesis() function in the brms package (Bürkner, 2017) which compares the prior and posterior densities of the interaction parameter at a specified value to check whether the tested effect is statically significant (Vuorre, 2017, March 21). As specified in the Savage-Dickey method (Wagenmakers et al., 2010) the hypothesis() function does not require a separate model for the null hypothesis since it instead uses the prior and posterior density to compute the Bayes factor. To do this, it samples posterior samples for the interaction term from the Bayesian model to represent the posterior distribution and then calculates the posterior density at the null value. Since the assumed null hypothesis is no interaction between memory load and semantic interference, the null hypothesis value for the

interaction is set at zero. Then the computed posterior density is compared to the prior density at the null value and using the Savage-Dickey density ratio, the Bayes factor is computed. The resulting Bayes factors for the hypothesis testing are summarised in Table 7 and are visualised in Figure 2.

Prior	**Bayes Factor**
Highly informative	10.77
Informative	63.48
Moderately informative	57.88
Broad	23.49

Table 6: Bayes factors for the different prior specifications for a sample size of 120 participants for the method of hypothesis testing.

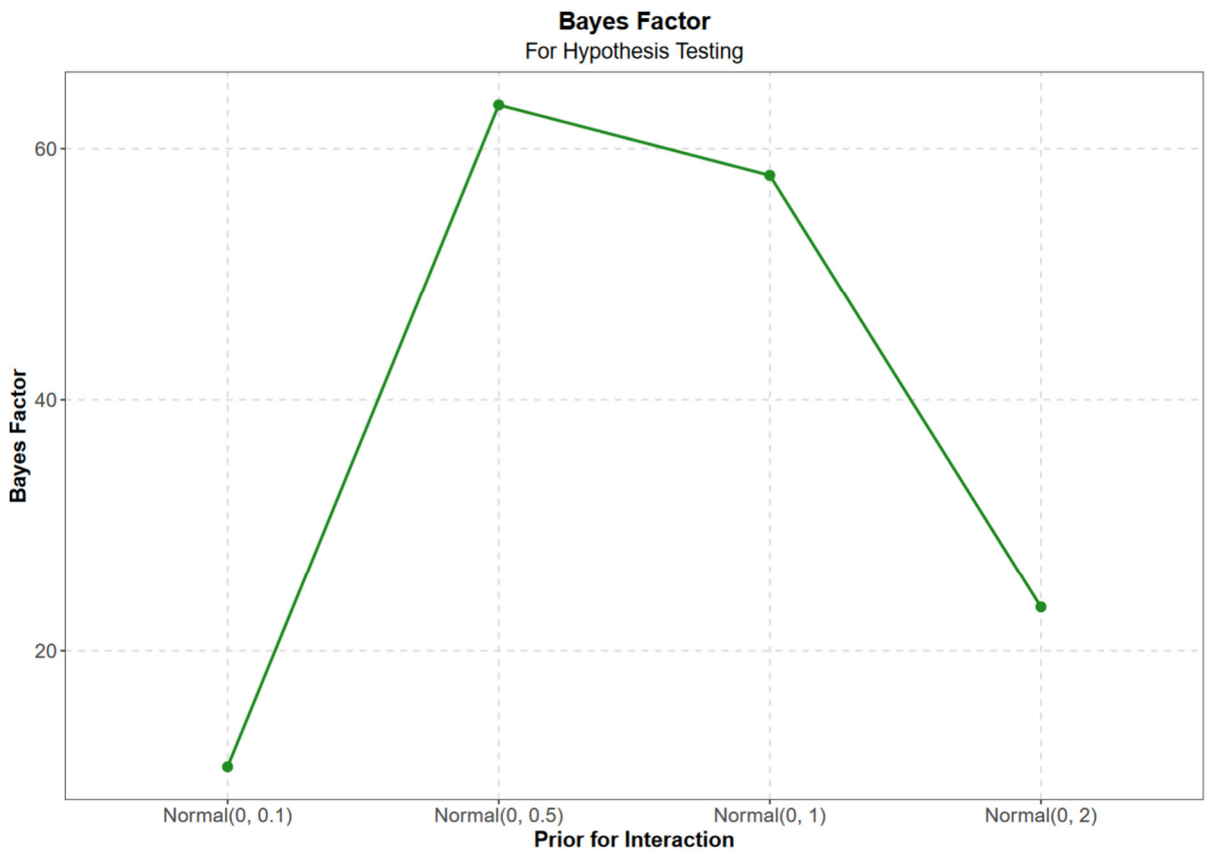

Figure 2: The Bayes factors for the interaction effect, resulting from the hypothesis testing for the different priors.

The informative prior had the highest Bayes factor (63.48), followed by the moderately informative prior (57.88). The broad prior returned a lower value (23.49) and the highly

informative prior had the lowest Bayes factor (10.77). These results show the influence that prior specification can have on the strength of evidence for the tested hypothesis. Nevertheless, all priors result in a Bayes factor that favours the alternative hypothesis over the null hypothesis and therefore makes it possible to conclude that an interaction effect can indeed be observed. In the next section, bridge sampling will be used to calculate the Bayes factor, following the suggestion by Nicenboim et al. (2024) to use more than one Bayesian method to check the stability of the estimates.

9.1.2 Bridge sampling

Bridge sampling uses the comparison of marginal likelihoods of the null and alternative model to calculate the Bayes factor. The marginal likelihood describes the probability of observing data under the assumption of a specific model with all the possible values of the parameters of the model included. This helps in assessing how well a model explains the data while still considering the uncertainty of the parameters of the model. Since it is challenging to calculate the marginal likelihood exactly, approximation methods such as bridge sampling are used (Gronau et al., 2017). Bridge sampling is useful when the computation of the marginal likelihood is time-consuming and computationally demanding which is the case with complex models such as the current model (Gronau et al., 2020).

In this thesis, bridge sampling is used to calculate approximately the marginal likelihood for each of the fitted Bayesian models with the different priors. To approximate the likelihood ratio between the two models, bridge sampling creates a so-called bridge-distribution. This bridge-distribution creates a bridge between the posterior distribution and random distributions that are simplified approximations of the posterior distribution and creates a more stable and efficient method for estimating the marginal likelihood compared to other MCMC methods such as the Savage-Dickey method (Nicenboim et al., 2024). The *bridgesampling* package in R (Gronau et al., 2020) was used to carry out the analysis. The marginal likelihoods for the full and null model were computed at several points within the posterior distribution and then were aggregated to provide an overall estimate for the marginal likelihood for each model. Once both marginal likelihoods were calculated, the Bayes factor was be computed by taking the ratio of those

likelihoods. For all priors, the Bayes factor was computed twice because bridge sampling uses an approximation method for likelihood estimation and as recommended in Nicenboim et al. (2024) sensitivity analyses should be run at least twice, even with a large sample size to ensure robust and stable estimates. This is particularly important because the bridge sampling algorithm relies on MCMC sampling, which introduces some randomness.

The results of the bridge sampling method indicate that, regardless of the prior used, all priors provide strong evidence in favour of the full model. The specific values are presented in Table 8 and are visualised in Figure 3. The least strong evidence for the full model is observed with the highly informative prior, probably due to its tightly constrained range and distribution that is centred closely around the specific parameter values. This reduces the flexibility of the model by limiting the parameter space.

Prior	**Bayes factor, first run**	**Bayes factor, second run**
Highly informative	10.64	10.96
Informative	79.90	80.30
Moderately informative	48.21	48.92
Broad	25.07	25.43

Table 7: Bayes factors for the different prior specifications for a sample size of 120 participants for the bridge sampling method, rounded to two decimals.

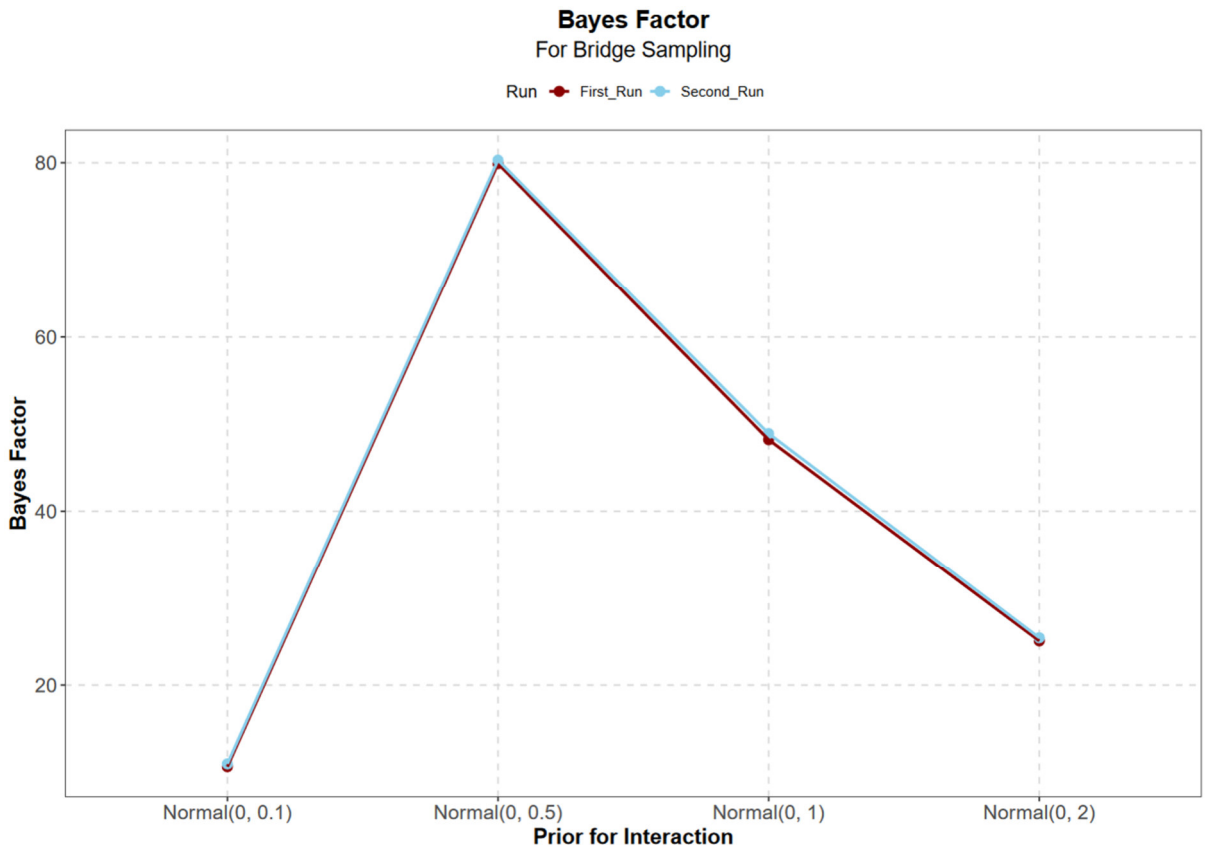

Figure 3: The Bayes factors for the interaction effect, resulting from the bridge sampling for the different priors. Two runs are compared: First_Run (red) and Second_Run (blue). Due to the lines being almost on top each other, it indicates indicate high consistency between the two runs.

The strongest evidence for the full model can be observed for the informative prior which indicates that the parameter space defined by this prior and the probability density around the specified values seem to align well with the posterior data. This results in a higher Bayes factor because the prior strongly supports the observed data. Unlike the highly informative prior which seems to constrains the parameter space too narrowly, the informative prior seems to balance the parameter space by neither being overly restrictive nor overly broad, as seen in the moderately and broad informative priors. The moderately and broad informative priors exhibit lower Bayes factors probably due to their wider parameter ranges which distribute the probability density across a broader spectrum. This reduces the concentration of probability in regions that are closely aligned with the posterior data and brings the focus to parameter values that are less compatible with the observed data and therefore lower the Bayes factor.

As can be seen in Table 7 and 8 the sensitivity analyses of both methods returned similar results which indicate that used methods are reliable and both showed consistent evidence for the Bayes factor supporting the full model.

9.2 Analysis of the ERPs

This section provides a brief recommendation on handling collected EEG data. However, as no actual data was collected for this thesis, the section is intentionally concise.

To be able to analyse the collected EEG data it is important to preprocess the data to eliminate potential confounds which includes filtering the EEG data to reduce noise and artifacts and to ensure smoother data processing (Niedermeyer, 2015). Additionally, participants displaying abnormal ERPs would need to be excluded to maintain data integrity. For the analysis, following the procedure outlined by Schoknecht et al. (2023), the activity of the centro-parietal electrodes should be averaged for each condition within the N400 time window (250–500 ms post-stimulus onset) as this is the region where the N400 component typically can be observed (Kutas & Federmeier, 2000). The ERP analysis should focus on the region of the critical verb, as well as the two preceding regions, since these might provide insights into the type of interference process occurring. Given that the critical region is located at the end of the sentence, posterior regions cannot be analysed.

If an interference effect is observed, it is important to identify the region of the sentence where the N400 effect appears, because this could provide an understanding about the nature of the interference. For instance, if an effect is found in the pre-critical region, it can be more likely associated with encoding processes, as retrieval has not yet occurred. But, if the effect is observed only in the critical region and potentially in the post-critical or spillover region, this might provide evidence for retrieval-based interference.

In the simulated data, an N400 effect is clearly observable and it was designed to be detected in the critical region to replicate the presence of an interference effect that can be attributed to retrieval interference. Table 9 shows the mean amplitudes observed across the four experimental conditions for the interaction parameter. The results show that the N400 amplitude is most negative in the high-interference condition with memory load (-

0.83 µV) which could reflect the increased semantic processing difficulty. The low-interference condition with memory load shows the least negative amplitude (-0.16 µV). The no-load baseline conditions show intermediate effects with high interference (-0.38 µV) and low interference (-0.50 µV) reflecting moderate levels of processing demand. Figure 4 complements Table 9 by visually illustrating the differences in N400 amplitudes between conditions with and without memory load. The figure highlights the interaction between semantic interference and memory load and clearly shows the contrast in amplitudes across the conditions. Together, Table 9 and Figure 4 provide an overview of how memory load and interference influence the N400 amplitudes in the simulated data which is consistent with the predictions of retrieval interference models.

Memory	**Interference**	**Mean Amplitude (in µV)**
Load	High Interference	-0.83
Load	Low Interference	-0.16
No Load	High Interference	-0.38
No Load	Low Interference	-0.50

Table 8: Mean N400 amplitudes for the interaction parameter in the different conditions, rounded to two decimals.

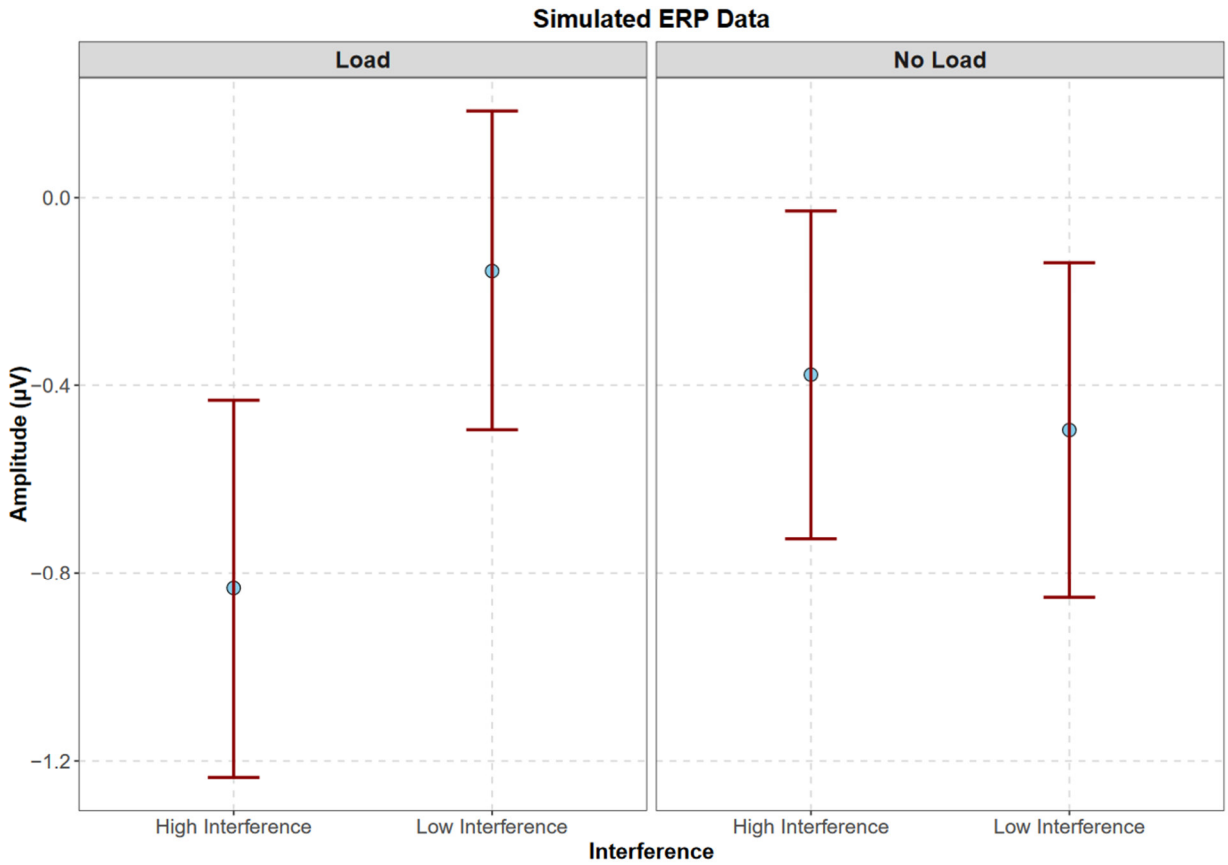

Figure 4: An overview of the mean N400 amplitudes for the interaction parameter in the Load and No Load condition, subdivided into the High and Low Interference condition. The blue dot indicates the mean value and the red error bars indicate ± 2 standard errors from the mean.

9.3 Comprehension question accuracy

After completing the study, it is important to also assess the accuracy of the participants on the comprehension questions to ensure the reliability of the data. Half of the experimental items were followed by a comprehension question and it could be possible that the memory items influence the comprehension accuracy and therefore, this relationship should also be examined. Participants with low accuracy rates should be excluded from analysis to maintain data quality. Previous studies (Mertzen et al., 2024; Schoknecht et al., 2023; Schoknecht et al., 2022; Van Dyke & McElree, 2006) report comprehension accuracies of 83-91% across conditions with the mean value being at 86%. No accuracy values below 80% have been reported. Hence, an exclusion threshold of participants that show accuracy below 80% on the comprehension questions seems to be a reasonable criterion. It would be recommended to analyse comprehension question

accuracies by condition to identify any potential impact of memory items on the performance.

9.4 Recall rate

The recall rate for the memory items should also be reported. Previous studies by Van Dyke and McElree (2006) and Mertzen et al. (2024) assessed the recall rates by using strict and lenient criteria. The strict criterion required all three sentence-external memory items to be recalled in the correct order whereas the lenient criterion only required the correct words regardless of the order. The percentages reported here are combined percentages of both studies. For the strict criterion the combined recall rates were 71% for the high interference condition and 73% for the low interference condition, with Mertzen et al. (2024) reporting lower rates in general. For the lenient criterion recall rates were 84% in the high interference condition and 86% in the low interference condition with Mertzen et al. (2024)'s rates being higher overall.

Mertzen et al. (2024) noted that some participants had extremely low recall rates, especially under the strict criterion with some as low as 0%. If similar results appear in future experiments, it could be advisable to conduct separate analyses including participants with very low recall in one and excluding those in another. For the strict criterion, a threshold of 60% could be applied to ensure that participants retained the sentence-external items in memory for the majority of the items since this is important for detecting possible interference effects. For the lenient criterion, a stricter threshold such as 70–80% might be more appropriate. These thresholds are recommendations and should be adjusted based on the observed data.

10. Discussion

This thesis used an EEG simulation study to investigate the phenomenon of semantic interference combined with the influence of sentence-external memory items. The results showed that the amplitude of the N400 effect is the largest in the condition in which the memory item matches the semantic cues of the verb. In the simulated data, this seems to indicate difficulties in the semantic processing and can also be related to demands on the working memory (Kutas & Federmeier, 2000). This is of course just an assumption based on previous studies regarding semantic processing and sentence-external items.

The results of the simulated study point towards proactive interference due to the memory items that act as distractors being sentence-external and appearing before the target item. This means that the memory items already have been encountered before the target item could have been encoded or retrieved. This prior processing of the memory items still affects the retrieval of the target item as can be seen by the existence of a more negative N400 amplitude in the high load and high interference conditions. Since proactive interference tends to show relatively small effects when compared to retroactive interference (Jäger et al., 2017; Van Dyke, 2011), it might be possible that the observed N400 amplitudes are not as pronounced in experimental data as they are in the simulated data.

10.1 Encoding versus cue-based retrieval interference

The design of the proposed study aims to distinguish between encoding and retrieval interference by structuring the experiment in a way that allows to attribute the observed effects to one of these processes. As highlighted by Schoknecht et al. (2023) when analysing the data it is important to focus on the critical region and the two preceding regions to determine in which region the interference effect (N400) can be observed. In the simulated study, it is assumed that the N400 effect will be observed in the critical region and not in pre-critical regions. When aiming to validate this assumption in a real experimental setting, the ERPs of the two preceding regions should also be analysed to confirm the absence of an N400 effect prior to the critical region.

Villata et al. (2018) noted that the design in Van Dyke & McElree (2006), which the current study is based on, ensures that the memory load remains constant across the conditions with only the semantic retrieval cues at the verb being manipulated. Therefore, if an interference effect in the critical region is observed only in the high-load condition, it can be attributed to semantic retrieval interference since at this stage of processing the semantic match between the memory words and the verb of the sentence becomes relevant. The interference effect appears when the previously encoded memory words are retrieved from memory to integrate them with the sentence meaning and the retrieval proves to be challenging or conflicting due to the semantic cue overlap with the subject of the sentence. The attempt of solving the competition between the memory words and the subject of the sentence and the resulting reanalysis of the sentence structure result in the interference effect. The attribution of this effect to retrieval interference could also be true if an interference effect would be observed in a possible spillover region.

In contrast, for encoding interference, the semantic match between the memory words and the verb is less relevant and the interference happens during earlier stages of sentence processing. Encoding interference mainly appears because of general cognitive load and competition for resources during the initial processing of memory words and sentence information rather than specific semantic relationships (Laurinavichyute et al., 2017). Encoding interference is based on the capacity limitations of the working memory that can interfere with the simultaneous processing of multiple pieces of information, such as keeping the memory words in memory while reading the ongoing sentence. When the working memory resources are overloaded, the processing slows down which results in a noticeable interference effect. Therefore, an N400 effect that can be observed in the pre-critical regions would suggest interference during the encoding phase because the memory words might compete with the sentence content during early processing. This type of interference is expected to appear regardless of whether the memory words semantically match the sentence since encoding interference is not tied to semantic processing but rather to the process of storing the words. This difference regarding the usage of semantic cues shows the different mechanisms underlying encoding and retrieval interference (Laurinavichyute et al., 2017). By examining the onset of the N400 effect in relation to the sentence structure and memory word presentation in the proposed study, it is possible to conclude whether the interference effect can be attributed mainly to affect encoding (initial memory processing) or retrieval (integration with sentence meaning).

In the simulated study the largest N400 amplitude was observed in the condition where the memory items semantically matched the retrieval cues of the verb and additionally, the N400 effect appears in the critical region which strongly supports the assumption that the interference is retrieval- rather than encoding-based. If the interference effect was due to encoding interference it would likely appear in the pre-critical regions. However, the absence of an N400 effect in these earlier regions and its appearance in the critical region points towards the role of semantic retrieval cues in the interference effect. The data supports the predictions of retrieval interference theories insofar that the interference occurs during the process of retrieving previously encoded memory items to integrate them with ongoing sentence processing. Then, the semantic overlap between the memory items and the verb creates competition that leads to retrieval difficulties and the observed N400 effect. These findings suggest that the interference observed in the simulated data can mainly be attributed to cue-based retrieval mechanisms rather than encoding processes.

Within the cue-based retrieval framework two main models of retrieval processes have been proposed: the Activation-Based Model (Lewis & Vasishth, 2005) and the Direct-Access Model (McElree, 2000). While both models agree that retrieval cues facilitate the access to memory items, they differ in their assumptions about the underlying mechanisms of interference.

The Activation-Based Model (ABM) as described in Nicenboim and Vasishth (2018) assumes that memory retrieval is based on a cue-driven activation mechanism. Words and phrases are conceptualised as combinations of lexical, semantic and syntactic features that serve as retrieval cues to identify the target item in memory. The items in memory are represented by activation levels with a higher activation indicating a better match to the retrieval cues. Those activation levels work in an additive way, meaning that the total activation of a memory item is determined by summing its baseline activation, cue-based activation (which is weighted by similarity to retrieval cues) and random noise. But, competition between items might happen and then retrieval difficulty might occur when multiple items in memory share similar retrieval cues which leads to simultaneous activation. This competition increases the likelihood of retrieving an incorrect distractor item and if no item matches the retrieval cues good enough to exceed the activation threshold, the retrieval fails completely. Another factor included in the model is decay

which describes that the activation of items decreases over time and therefore making older items more difficult to retrieve.

The Direct-Access Model (DAM), as described by Van Dyke and McElree (2006), mainly differs from the ABM in its rejection of competition between items based on activation levels but still shares several similarities with the ABM. Both models assume that words and phrases in memory are represented as feature bundles and that the degree of match between retrieval cues and memory items helps in distinguishing the target item from competitors. In the DAM, the retrieval cues are combined multiplicatively which means that the probability of retrieving an item is determined by the product of match strengths across all relevant cues. Both models agree that memory retrieval relies on the match between cues and the target item and the match can be reduced by the extent to which competing items also match the cues. As mentioned in Nicenboim and Vasishth (2018) an important distinction between those two models is the role of competition. In the DAM the overlap of cues with the memory representation allows direct access to the relevant memory items and therefore the likelihood of target retrieval can be affected by inhibitory interference, especially in high-interference conditions when several items match the retrieval cues. Contrarily, in the ABM the retrieval difficulty is based on the competition between items and the need to resolve this competition before having retrieval access to the selected item which would also be especially visible in the high-interference condition when several items match the retrieval cues.

The observed N400 effect in the simulated data shows support for both, the Activation-Based Model and the Direct-Access Model since both predict retrieval interference effects in memory. The ABM attributes the more negative N400 amplitude in high-interference conditions with memory load to the activation competition among items that share similar retrieval cues which then leads to increased difficulty in retrieving the target item. The DAM predicts as well that high interference reduces the probability of correctly retrieving the target item but bases this assumption on the reduced overlap between retrieval cues and memory representations which then also manifests as a more negative N400 amplitude. It seems that the simulated data aligns with the general retrieval interference prediction that is shared by both models and further evidence would be needed to be able to distinguish between the two models. If in future research an experimental study can replicate the data of the simulation and therefore support and

validate theories of cue-based retrieval interference, a follow-up study could be designed to determine which retrieval cue model offers a better explanation of the results or is more likely to be accurate.

Even though the simulated data seems to show evidence for cue-based retrieval interference rather than encoding interference, it does not mean that the two processes are exclusive or cannot interact. In Yadav et al. (2021) a computational analysis of encoding and retrieval processes has shown that encoding can modulate retrieval. Furthermore, in Yadav et al. (2023) the possibility is proposed that retrieval and encoding processes might take place simultaneously and complement each other, instead of being entirely different mechanisms. The simulated data in the current study focused on modelling cue-based retrieval interference effects because it was informed by findings from previous ERP studies on this topic. However, in Schoknecht et al. (2023) it is pointed out that the experimental procedure might be an important factor when determining which interference effects are observed. For instance, in their study in the self-paced reading data encoding interference was identified while in the ERP data retrieval interference was detected. This suggests that the method of stimulus presentation including the timing and control of presentation, might impact the observed effects. Schoknecht et al. (2023) assume that the automated presentation of sentence phrases with fixed timings in ERP studies might mask potential encoding effects. In contrast, the self-paced reading allows participants to adjust their reading speed according to their needs (particularly when encountering the distractors) and therefore making it possible for encoding interference to be observed. Building on Yadav et al. (2023) it appears that neither encoding interference nor cue-based retrieval interference models alone are enough to fully explain the findings in their study. This suggests the necessity of a hybrid model that includes the observed phenomena.

In the current simulated data only cue-based retrieval interference was modelled. To build upon the theoretical assumptions from the simulated data, the results must first be validated through a study conducted in an experimental setting. Afterwards, a follow-up study could be designed to investigate the potential presence of encoding interference effects in the ERPs. For this study it might be helpful to use a self-paced reading paradigm combined with EEG recordings to test whether encoding interference effects can appear alongside retrieval interference effects. While this approach will introduce additional

noise into the EEG data, it could help in isolating potential encoding interference effects together with retrieval interference effects.

10.2 Retrieval interference and memory

This study simulated EEG data to investigate the impact of sentence-external distractors on the N400 amplitude and the findings indicate that sentence-external distractors can influence the amplitude. This suggests that retrieval from semantic memory can be affected by competition between the target item and distractor items even when the distractors are located outside of the sentence context. Therefore, semantic memory is not only sensitive to linguistic input from within the sentence but also uses representations from external sources that are already stored in short-term memory to resolve a competition. Thus, the N400 might not only reflect semantic match or mismatch but also memory retrieval processes (as mentioned in Kutas & Federmeier, 2000, 2011; Schoknecht et al., 2023). The sentence-external distractors that must be held in working memory during the sentence processing because of the design of the experimental task can interfere with the sentence content to the extent that a change in the N400 amplitude is observed. This seems to suggest a connection between semantic memory, working memory and the N400. While the proposed effects need further validation through experimental studies, the assumed results seem to indicate that semantic memory integrates information from both internal and external sources when processing a sentence.

The possible connection of the N400 to working memory raises the question whether the working memory capacity of individuals could have an influence on the N400 amplitude and the interference effects in the proposed experimental design. It could be possible that participants with higher working memory capacity show reduced interference since they can better maintain and manipulate the competing linguistic information. On the contrary, it might be assumed that participants with lower working memory capacity then possibly show stronger N400 and interference effects. An effect of working memory on N400 amplitudes has been suggested by other studies such as Nakano et al. (2010) and could be further investigated by conducting the proposed study as an experimental EEG study with an included working memory assessment test (an overview of tests is offered in

Conway et al. (2005)). Future research could examine those potential individual differences in working memory to improve the generalisability of N400 findings across diverse participant groups.

10.3 Temporal and spatial considerations

When analysing ERPs it is important to consider the time of their appearance since an N400 should appear between 250 to 500ms (Kutas & Federmeier, 2000). Other negativities might also appear at the same electrodes but within different time windows and this temporal distinction helps to differentiate between various ERP components. Additionally, the specific locations of the electrodes where the ERP component has been recorded must be examined. In the simulated data it was assumed based on prior studies such as Schoknecht et al. (2023) that the negativity in the ERPs would be observed in the region of the centro-parietal electrodes. However, in experimental data it is important to verify which electrodes record the increased negativity because other interference effects might also produce negativities although at different electrodes from those associated with the N400 (Lee & Garnsey, 2015). For example, the Left Anterior Negativity (LAN) is another ERP component that can be associated with interference (Lee & Garnsey, 2015; Vasishth & Drenhaus, 2011) and is typically measured at the left-frontal electrodes and like the N400 it appears around 300 to 500 ms in the post-stimulus time window but it is usually more linked to morphosyntactic violations (Lee & Garnsey, 2015). Since the time window of appearance of the N400 and the LAN is highly overlapping it is important to check the location of appearance to avoid misinterpretation of the observed effect. However, since the manipulation in this study is semantic rather than syntactic, the LAN is unlikely to appear. But since it has also been suggested that the LAN can reflect an increased memory load during processing (Kluender & Kutas, 1993; Martín-Loeches et al., 2005) which would be the case in this study, in order to be able to correctly classify the observed ERP effect, it is necessary to check which specific electrodes recorded the negativity.

10.4 Comparison of Bayesian models

The results of the analysis based on the Bayes factor indicate that both testing methods, hypothesis testing and bridge sampling, support the full model over the null model and therefore support the finding of an interaction effect between semantic interference and memory load. However, it seems that sample size plays a very important role when it comes to the stability of the Bayes factor estimates since both methods showed anomalous values when being computed for 100 participants whereas the results were stable and interpretable for 120 participants. This further shows that choosing an appropriate sample size for an experiment is very important and a preliminary power analysis can assist in choosing the minimum sample size. Based on the power analysis for this thesis, the Bayesian analysis needs both the mean and the upper bound value to achieve at least 80% power to function properly. Furthermore, it can be seen that the prior specification impacts the outcome of the Bayes factor analysis quite strongly and thus points out the importance of choosing the right priors, based on previous studies, knowledge and preferably conducting a sensitivity analysis. The difference in Bayes factor for the hypothesis testing method and the bridge sampling method (for 120 participants) show that different Bayesian models can result in different outcomes although the direction of the outcome (favouring the full model over the null model) still remains identical. Since bridge sampling is computationally more intensive, it could be recommended to use the hypothesis testing methods when computational limitations are to be expected. Overall, both analyses of the simulated data provided evidence for an interaction between semantic interference and memory load and therefore support cue-based retrieval interference theories.

10.5 Limitations

The primary limitation of this study is that it only has been conducted as a simulation and needs yet to be tested in a laboratory setting with real participants. This would be necessary to investigate whether the assumptions made in this thesis that are based on prior research can be verified in an actual experimental context. One challenge in this might be the interpretation of the results from the real-life EEG data since the simulated data does not take into account the artifacts, noise distributions and brain activity that might be seen in actual EEG data. Also factors such as individual variability in eye blinks or muscle movements and individual baselines are not taken into account in the simulated data. Unlike in simulated data this noise necessities preprocessing during which it could be possible that significant amounts of data need be excluded which might impact the data points that will be analysed and could hide potential effects. But to ensure that the assumed findings can be validated experimentally and are robust, empirical EEG data is needed.

Additionally, the items and fillers that were proposed in this thesis are exclusively in German. As a result, any experimental findings that are derived from the items provided in this thesis would only allow to draw conclusions about the interference of sentence-external memory items in German. After successful implementation in German, it could be informative to expand the study to other languages to see whether the results can be generalised cross-linguistically. Similarly to Mertzen et al. (2024) a crosslinguistic study could be used to either support or provide evidence against the assumption of cue-based retrieval interference across several languages.

In the materials presented further above, it can be seen that the comprehension questions have to be answered with either an affirmative answer ("Yes") or a negative answer ("No"). In Nicenboim et al. (2018) it has been noted that Yes/No questions might lead to the strategy of good-enough processing (Ferreira et al., 2002) of the participants. This could be problematic since the participants expect a Yes/No- question and therefore might be tempted to use shallow parsing instead of paying close attention to the content of the sentence which could impact the amplitude of the N400 to the extent that no effect might be visible. While the items in this study should not be complex enough to necessitate good-enough processing, participants could still rely on this strategy due to fatigue or loss

of concentration. Since the purpose of this study was a replication of the study by Van Dyke and McElree (2006) in an EEG paradigm, an identical format of questions was used to stay as close to the original items as possible. It could be advisable to vary the question types in potential follow-up studies to avoid this confound.

Another possible confound might be the placement of the critical region at the end of the experimental item which could hide potential spillover effects. Mitchell (2018) discusses the possibility that positioning the critical region at the end of an experimental item might cause unfinished processing to spillover to the next task or trial. This could affect the measurements of the critical effect to such an extent that it will be observed in the next measurement or cannot be observed at all because of processing delays. In the context of the present study, this could mean that the expected negativity of the N400 amplitude might be observed while the participants are reading the comprehension question or even later. Alternatively, it is also possible that no effect can be observed since the processing was interrupted and then overridden by the cognitive demands of the comprehension question. However, the items were designed in this way in order to replicate the study by Van Dyke and McElree (2006) as closely as possible. In a future study this confound could be avoided by adding another phrase after the critical region.

11. Conclusion

The current study simulated EEG data to investigate the interaction between semantic interference and memory load in the context of sentence-external distractors with a particular focus on the N400 component as an indicator for retrieval interference. The simulated findings seem to indicate proactive interference and support the hypothesis that sentence-external distractors can impact semantic processing by observing a higher N400 negativity in the condition of high memory load and high interference due to a semantic match between the target and distractor items. Therefore, it seems that sentence-external distractors are able to disrupt the retrieval of the target item even if the target item has not been encountered yet. This observation supports the assumption of cue-based retrieval theories since the simulated data shows a pattern that has been predicted by memory retrieval theories such as the Activation-Based Model (Nicenboim & Vasishth, 2018) and the Direct-Access Model (Van Dyke & McElree, 2006). No evidence for encoding interference can be seen. However, this does not necessarily exclude the possibility of encoding interference happening simultaneously and future work could focus on further examining the potential role of encoding interference in this experimental design. Besides, the relationship between working memory capacity and the semantic memory could be investigated further.

Despite those findings, several limitations have been identified that would need to be addressed in future work. Firstly, the simulated results need to be verified in an empirical, laboratory-based study with real participants. Furthermore, it would be advisable to carry out the research in multiple languages to determine whether the findings can be generalised across languages. Besides, potentially adjusting the experimental design such as modifying the questions and the items slightly to avoid good-enough processing and concealment of observed effects could be advisable.

Overall, this study provides theoretical support for the assumptions of cue-based retrieval interference theories and expands those assumptions to also include the interference from sentence-external items.

References

Anderson, M. C., & Neely, J. H. (1996). Interference and Inhibition in Memory Retrieval. In *Memory* (pp. 237–313). Elsevier. https://doi.org/10.1016/B978-012102570-0/50010-0

Araya, R. (2014). Input transformation by dendritic spines of pyramidal neurons. *Frontiers in Neuroanatomy*, *8*, 141. https://doi.org/10.3389/fnana.2014.00141

Barrett, S. E., & Rugg, M. D. (1989). Event-related potentials and the semantic matching of faces. *Neuropsychologia*, *27*(7), 913–922. https://doi.org/10.1016/0028-3932(89)90067-5

Bates, D., Mächler, M., Bolker, B., & Walker, S. (2015). Fitting Linear Mixed-Effects Models Using lme4. *Journal of Statistical Software*, *67*(1). https://doi.org/10.18637/jss.v067.i01

Belletti, A., Friedmann, N., Brunato, D., & Rizzi, L. (2012). Does gender make a difference? Comparing the effect of gender on children's comprehension of relative clauses in Hebrew and Italian. *Lingua. International Review of General Linguistics. Revue Internationale De Linguistique Generale*, *122*(10), 1053–1069. https://doi.org/10.1016/j.lingua.2012.02.007

Beres, A. M. (2017). Time is of the Essence: A Review of Electroencephalography (EEG) and Event-Related Brain Potentials (ERPs) in Language Research. *Applied Psychophysiology and Biofeedback*, *42*(4), 247–255. https://doi.org/10.1007/s10484-017-9371-3

Betancourt, M. (2017, January 10). *A Conceptual Introduction to Hamiltonian Monte Carlo*. http://arxiv.org/pdf/1701.02434v2

Brown, J. (1958). Some Tests of the Decay Theory of Immediate Memory. *Quarterly Journal of Experimental Psychology*, *10*(1), 12–21. https://doi.org/10.1080/17470215808416249

Bürkner, P.-C. (2017). brms : An R Package for Bayesian Multilevel Models Using Stan. *Journal of Statistical Software*, *80*(1). https://doi.org/10.18637/jss.v080.i01

Campanelli, L., Van Dyke, J. A., & Marton, K. (2018). The Modulatory Effect of Expectations on Memory Retrieval During Sentence Comprehension. *Proceedings of the 40th Annual Meeting of the Cognitive Science Society*, 1434–1439.

Carpenter, B., Gelman, A., Hoffman, M. D., Lee, D., Goodrich, B., Betancourt, M., Brubaker, M. A., Guo, J., Li, P., & Riddell, A. (2017). Stan: A Probabilistic Programming Language. *Journal of Statistical Software*, *76.* https://doi.org/10.18637/jss.v076.i01

Carpenter, P. A., Just, M. A., & Shell, P. (1990). What one intelligence test measures: A theoretical account of the processing in the Raven Progressive Matrices Test. *Psychological Review*, *97*(3), 404–431. https://doi.org/10.1037/0033-295X.97.3.404

Chomsky, N. (Ed.). (1996). *Studies on Semantics in Generative Grammar*. DE GRUYTER. https://doi.org/10.1515/9783110814231

Conway, A. R. A., Kane, M. J., Bunting, M. F., Hambrick, D. Z., Wilhelm, O [Oliver], & Engle, R. W. (2005). Working memory span tasks: A methodological review and user's guide. *Psychonomic Bulletin & Review*, *12*(5), 769–786. https://doi.org/10.3758/BF03196772

Cowan, N. (2001). The magical number 4 in short-term memory: A reconsideration of mental storage capacity. *The Behavioral and Brain Sciences*, *24*(1), 87-114; discussion 114-85. https://doi.org/10.1017/S0140525X01003922

Crowder, R. G. (2014). *Principles of Learning and Memory*. Psychology Press. https://doi.org/10.4324/9781315746944

Engelmann, F., Jäger, L. A., & Vasishth, S. (2019). The Effect of Prominence and Cue Association on Retrieval Processes: A Computational Account. *Cognitive Science*, *43*(12), e12800. https://doi.org/10.1111/cogs.12800

Federmeier, K. D., & Kutas, M [Marta] (1999). A Rose by Any Other Name: Long-Term Memory Structure and Sentence Processing. *Journal of Memory and Language*, *41*(4), 469–495. https://doi.org/10.1006/jmla.1999.2660

Federmeier, K. D., & Kutas, M [Marta] (2001). Meaning and modality: Influences of context, semantic memory organization, and perceptual predictability on picture processing. *Journal of Experimental Psychology. Learning, Memory, and Cognition*, *27*(1), 202–224. https://doi.org/10.1037/0278-7393.27.1.202

Fedorenko, E., Gibson, E., & Rohde, D. (2006). The nature of working memory capacity in sentence comprehension: Evidence against domain-specific working memory resources☆. *Journal of Memory and Language*, *54*(4), 541–553. https://doi.org/10.1016/j.jml.2005.12.006

Ferreira, F., Bailey, K. G., & Ferraro, V. (2002). Good-Enough Representations in Language Comprehension. *Current Directions in Psychological Science*, *11*(1), 11–15. https://doi.org/10.1111/1467-8721.00158

Friederici, A. D., & Thierry, G. (2008). *Early Language Development* (Vol. 5). John Benjamins Publishing Company. https://doi.org/10.1075/tilar.5

Gamerman, D., & Lopes, H. F. (2006). *Markov Chain Monte Carlo*. Chapman and Hall/CRC. https://doi.org/10.1201/9781482296426

Gaskell, M. G. (Ed.). (2007). *The Oxford handbook of psycholinguistics*. Oxford University Press.

Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., & Rubin, D. B. (2013). *Bayesian Data Analysis*. Chapman and Hall/CRC. https://doi.org/10.1201/b16018

Gordon, P. C., Hendrick, R [R.], & Johnson, M [M.] (2001). Memory interference during language processing. *Journal of Experimental Psychology. Learning, Memory, and Cognition*, *27*(6), 1411–1423. https://doi.org/10.1037//0278-7393.27.6.1411

Gordon, P. C., Hendrick, R [Randall], & Johnson, M [Marcus] (2004). Effects of noun phrase type on sentence complexity. *Journal of Memory and Language*, *51*(1), 97–114. https://doi.org/10.1016/j.jml.2004.02.003

Gordon, P. C., Hendrick, R [Randall], & Levine, W. H. (2002). Memory-load interference in syntactic processing. *Psychological Science*, *13*(5), 425–430. https://doi.org/10.1111/1467-9280.00475

Gronau, Q. F., Sarafoglou, A., Matzke, D., Ly, A., Boehm, U., Marsman, M., Leslie, D. S., Forster, J. J., Wagenmakers, E.-J., & Steingroever, H. (2017). A tutorial on bridge sampling. *Journal of Mathematical Psychology*, *81*, 80–97. https://doi.org/10.1016/j.jmp.2017.09.005

Gronau, Q. F., Singmann, H., & Wagenmakers, E.-J. (2020). bridgesampling : An R Package for Estimating Normalizing Constants. *Journal of Statistical Software*, *92*(10). https://doi.org/10.18637/jss.v092.i10

Hoffman, M., & Gelman, A. (2014). The No-U-Turn Sampler: Adaptively Setting Path Lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*(15), 1351–1381.

Hofmeister, P., & Vasishth, S. (2014). Distinctiveness and encoding effects in online sentence comprehension. *Frontiers in Psychology*, *5*, 1237. https://doi.org/10.3389/fpsyg.2014.01237

Holtmann, J., Koch, T., Lochner, K., & Eid, M. (2016). A Comparison of ML, WLSMV, and Bayesian Methods for Multilevel Structural Equation Models in Small Samples: A Simulation Study. *Multivariate Behavioral Research*, *51*(5), 661–680. https://doi.org/10.1080/00273171.2016.1208074

Husain, S., Vasishth, S., & Srinivasan, N. (2014). Strong expectations cancel locality effects: Evidence from Hindi. *PloS One*, *9*(7), e100986. https://doi.org/10.1371/journal.pone.0100986

Jäger, L. A., Benz, L., Roeser, J., Dillon, B. W., & Vasishth, S. (2015). Teasing apart retrieval and encoding interference in the processing of anaphors. *Frontiers in Psychology*, *6*, 506. https://doi.org/10.3389/fpsyg.2015.00506

Jäger, L. A., Engelmann, F., & Vasishth, S. (2017). Similarity-based interference in sentence comprehension: Literature review and Bayesian meta-analysis. *Journal of Memory and Language*, *94*, 316–339. https://doi.org/10.1016/j.jml.2017.01.004

Jäger, L. A., Mertzen, D., Van Dyke, J. A., & Vasishth, S. (2020). Interference patterns in subject-verb agreement and reflexives revisited: A large-sample study. *Journal of Memory and Language*, *111*. https://doi.org/10.1016/j.jml.2019.104063

Jeffreys, H. (1939). *The theory of probability*. Clarendon Pr.

Jeffreys, H. (2003). *Theory of probability* (3. ed., repr). *Oxford classic texts in the physical sciences*. Clarendon Pr.

Klimesch, W. (1999). Eeg alpha and theta oscillations reflect cognitive and memory performance: A review and analysis. *Brain Research. Brain Research Reviews*, *29*(2-3), 169–195. https://doi.org/10.1016/S0165-0173(98)00056-3

Kluender, R., & Kutas, M [M.] (1993). Bridging the Gap: Evidence from ERPs on the Processing of Unbounded Dependencies. *Journal of Cognitive Neuroscience*, *5*(2), 196–214. https://doi.org/10.1162/jocn.1993.5.2.196

Kutas, M [Marta], & Federmeier, K. D. (2000). Electrophysiology reveals semantic memory use in language comprehension. *Trends in Cognitive Sciences*, *4*(12), 463–470. https://doi.org/10.1016/s1364-6613(00)01560-6

Kutas, M [Marta], & Federmeier, K. D. (2011). Thirty years and counting: Finding meaning in the N400 component of the event-related brain potential (ERP).

Annual Review of Psychology, *62*, 621–647. https://doi.org/10.1146/annurev.psych.093008.131123

Kutas, M [Marta], & Hillyard, S. A. (1980). Reading senseless sentences: Brain potentials reflect semantic incongruity. *Science (New York, N.Y.)*, *207*(4427), 203–205. https://doi.org/10.1126/science.7350657

Kutas, M [Marta], & Hillyard, S. A. (1984). Brain potentials during reading reflect word expectancy and semantic association. *Nature*, *307*(5947), 161–163. https://doi.org/10.1038/307161a0

Laurinavichyute, A., Jäger, L. A., Akinina, Y., Roß, J., & Dragoy, O. (2017). Retrieval and Encoding Interference: Cross-Linguistic Evidence from Anaphor Processing. *Frontiers in Psychology*, *8*, 965. https://doi.org/10.3389/fpsyg.2017.00965

Lee, E.-K., & Garnsey, S. M. (2015). An ERP study of plural attraction in attachment ambiguity resolution: Evidence for retrieval interference. *Journal of Neurolinguistics*, *36*, 1–16. https://doi.org/10.1016/j.jneuroling.2015.04.004

Lewis, R. L., & Vasishth, S. (2005). An activation-based model of sentence processing as skilled memory retrieval. *Cognitive Science*, *29*(3), 375–419. https://doi.org/10.1207/s15516709cog0000_25

Lewis, R. L., Vasishth, S., & Van Dyke, J. A. (2006). Computational principles of working memory in sentence comprehension. *Trends in Cognitive Sciences*, *10*(10), 447–454. https://doi.org/10.1016/j.tics.2006.08.007

Luck, S. J. (2014). *An introduction to the event-related potential technique* (Second edition). *A Bradford book*. The MIT Press.

Martin, A. E. (2018). Cue integration during sentence comprehension: Electrophysiological evidence from ellipsis. *PloS One*, *13*(11), e0206616. https://doi.org/10.1371/journal.pone.0206616

Martin, A. E., & McElree, B [Brian] (2009). Memory operations that support language comprehension: Evidence from verb-phrase ellipsis. *Journal of Experimental Psychology. Learning, Memory, and Cognition*, *35*(5), 1231–1239. https://doi.org/10.1037/a0016271

Martin, A. E., Nieuwland, M. S., & Carreiras, M. (2012). Event-related brain potentials index cue-based retrieval interference during sentence comprehension. *NeuroImage*, *59*(2), 1859–1869. https://doi.org/10.1016/j.neuroimage.2011.08.057

Martin, A. E., Nieuwland, M. S., & Carreiras, M. (2014). Agreement attraction during comprehension of grammatical sentences: Erp evidence from ellipsis. *Brain and Language*, *135*, 42–51. https://doi.org/10.1016/j.bandl.2014.05.001

Martín-Loeches, M., Muñoz, F., Casado, P., Melcón, A., & Fernández-Frías, C. (2005). Are the anterior negativities to grammatical violations indexing working memory? *Psychophysiology*, *42*(5), 508–519. https://doi.org/10.1111/j.1469-8986.2005.00308.x

McElree, B [B.] (2000). Sentence comprehension is mediated by content-addressable memory structures. *Journal of Psycholinguistic Research*, *29*(2), 111–123. https://doi.org/10.1023/A:1005184709695

McPherson, W. B., & Holcomb, P. J. (1999). An electrophysiological investigation of semantic priming with pictures of real objects. *Psychophysiology*, *36*(1), 53–65. https://doi.org/10.1017/S0048577299971196

Mertzen, D., Laurinavichyute, A., Dillon, B. W., Engbert, R., & Vasishth, S. (2024). Crosslinguistic evidence against interference from extra-sentential distractors. *Journal of Memory and Language*, *137*, 104514. https://doi.org/10.1016/j.jml.2024.104514

Mitchell, D. C. (2018). An evaluation of subject-paced reading tasks and other methods of investigating immediate processes in reading. In M. A. Just & D. E. Kieras (Eds.), *Routledge Revivals. New Methods in Reading Comprehension Research* (First edition). Taylor and Francis.

Morris, W. K., Vesk, P. A., McCarthy, M. A., Bunyavejchewin, S., & Baker, P. J. (2015). The neglected tool in the Bayesian ecologist's shed: A case study testing informative priors' effect on model accuracy. *Ecology and Evolution*, *5*(1), 102–108. https://doi.org/10.1002/ece3.1346

Nairne, J. S. (1990). A feature model of immediate memory. *Memory & Cognition*, *18*(3), 251–269. https://doi.org/10.3758/BF03213879

Nairne, J. S. (2002a). The myth of the encoding-retrieval match. *Memory (Hove, England)*, *10*(5-6), 389–395. https://doi.org/10.1080/09658210244000216

Nairne, J. S. (2002b). Remembering over the short-term: The case against the standard model. *Annual Review of Psychology*, *53*, 53–81. https://doi.org/10.1146/annurev.psych.53.100901.135131

Nakano, H., Saron, C., & Swaab, T. Y. (2010). Speech and span: Working memory capacity impacts the use of animacy but not of world knowledge during spoken sentence comprehension. *Journal of Cognitive Neuroscience*, *22*(12), 2886–2898. https://doi.org/10.1162/jocn.2009.21400

Nicenboim, B., Schad, D. J., & Vasishth, S. (2024). *An Introduction to Bayesian Data Analysis for Cognitive Science*. https://bruno.nicenboim.me/bayescogsci/

Nicenboim, B., & Vasishth, S. (2018). Models of retrieval in sentence comprehension: A computational evaluation using Bayesian hierarchical modeling. *Journal of Memory and Language*, *99*, 1–34. https://doi.org/10.1016/j.jml.2017.08.004

Nicenboim, B., Vasishth, S., Engelmann, F., & Suckow, K. (2018). Exploratory and Confirmatory Analyses in Sentence Processing: A Case Study of Number Interference in German. *Cognitive Science*, *42 Suppl 4*, 1075–1100. https://doi.org/10.1111/cogs.12589

Niedermeyer, E. (2015). *Electroencephalography: Basic Principles, Clinical Applications, and Related Fields*. Wolters Kluwer. http://gbv.eblib.com/patron/FullRecord.aspx?p=3418225

Oberauer, K., Süß, H.-M., Schulze, R., Wilhelm, O [O.], & Wittmann, W. W. (2000). Working memory capacity — facets of a cognitive ability construct. *Personality and Individual Differences*, *29*(6), 1017–1045. https://doi.org/10.1016/S0191-8869(99)00251-2

Öztekin, I., & McElree, B [Brian] (2007). Proactive interference slows recognition by eliminating fast assessments of familiarity. *Journal of Memory and Language*, *57*(1), 126–149. https://doi.org/10.1016/j.jml.2006.08.011

Petten, C. V., Kutas, M [Marta], Kluender, R., Mitchiner, M., & McIsaac, H. (1991). Fractionating the word repetition effect with event-related potentials. *Journal of Cognitive Neuroscience*, *3*(2), 131–150. https://doi.org/10.1162/jocn.1991.3.2.131

Purves, D., & Williams, S. M. (Eds.). (2001). *Neuroscience* (2nd ed.). Sinauer Associates.

R Core Team. (2024). *R: A Language and Environment for Statistical Computing* [Computer software]. R Foundation for Statistical Computing. https://www.R-project.org/

Rabe, M. M., Paape, D., Mertzen, D., Vasishth, S., & Engbert, R. (2024). SEAM: An integrated activation-coupled model of sentence processing and eye movements in reading. *Journal of Memory and Language*, *135*, 104496. https://doi.org/10.1016/j.jml.2023.104496

Reis, D. J., Kaizer, A. M., Kinney, A. R., Bahraini, N. H., Holliday, R., Forster, J. E., & Brenner, L. A. (2023). A practical guide to random-effects Bayesian meta-analyses with application to the psychological trauma and suicide literature. *Psychological Trauma : Theory, Research, Practice and Policy*, *15*(1), 121–130. https://doi.org/10.1037/tra0001316

Sarfarazi, M., Cave, B., Richardson, A., Behan, J., & Sedgwick, E. M. (1999). Visual event related potentials modulated by contextually relevant and irrelevant olfactory primes. *Chemical Senses*, *24*(2), 145–154. https://doi.org/10.1093/chemse/24.2.145

Schad, D. J., Nicenboim, B., Bürkner, P.-C., Betancourt, M., & Vasishth, S. (2021). *Workflow Techniques for the Robust Use of Bayes Factors.* https://doi.org/10.48550/arXiv.2103.08744

Schoknecht, P., Himanshu, Y., & Vasishth, S. (2023). Do syntactic and semantic similarity lead to interference effects? Evidence from self-paced reading and event-related potentials using German. Advance online publication. https://doi.org/10.31234/osf.io/cwymg

Schoknecht, P., Roehm, D., Schlesewsky, M., & Bornkessel-Schlesewsky, I. (2022). The interaction of predictive processing and similarity-based retrieval interference: an ERP study. *Language, Cognition and Neuroscience*, *37*(7), 883–901. https://doi.org/10.1080/23273798.2022.2026421

Schomer, D. L., & Da Lopes Silva, F. H. (Eds.). (2017). *Niedermeyer's electroencephalography: Basic principles, clinical applications, and related fields* (Seventh edition). Oxford University Press. https://doi.org/10.1093/med/9780190228484.001.0001

Schützler, O., & Schlüter, J. (2022). *Data and Methods in Corpus Linguistics*. Cambridge University Press. https://doi.org/10.1017/9781108589314

Smith, G., Franck, J., & Tabor, W. (2021). Encoding interference effects support self-organized sentence processing. *Cognitive Psychology*, *124*, 101356. https://doi.org/10.1016/j.cogpsych.2020.101356

Stan Development Team. (2024). *RStan: the R interface to Stan* (Version R package version 2.32.6) [Computer software]. https://mc-stan.org/

Tanner, D., Nicol, J., & Brehm, L. (2014). The time-course of feature interference in agreement comprehension: Multiple mechanisms and asymmetrical attraction.

Journal of Memory and Language, *76*, 195–215. https://doi.org/10.1016/j.jml.2014.07.003

TEMoore. ((n.d.)). *Bayesian Regression Analysis in R using brms*. https://tem11010.github.io/regression_brms/

Van Dyke, J. A. (2003). Distinguishing effects of structure and decay on attachment and repair: A cue-based parsing account of recovery from misanalyzed ambiguities. *Journal of Memory and Language*, *49*(3), 285–316. https://doi.org/10.1016/S0749-596X(03)00081-0

Van Dyke, J. A. (2007). Interference effects from grammatically unavailable constituents during sentence processing. *Journal of Experimental Psychology. Learning, Memory, and Cognition*, *33*(2), 407–430. https://doi.org/10.1037/0278-7393.33.2.407

Van Dyke, J. A. (2011). Cue-dependent interference in comprehension. *Journal of Memory and Language*, *65*(3), 247–263. https://doi.org/10.1016/j.jml.2011.05.002

Van Dyke, J. A., Johns, C. L., & Kukona, A. (2014). Low working memory capacity is only spuriously related to poor reading comprehension. *Cognition*, *131*(3), 373–403. https://doi.org/10.1016/j.cognition.2014.01.007

Van Dyke, J. A., & McElree, B [Brian] (2006). Retrieval interference in sentence comprehension. *Journal of Memory and Language*, *55*(2), 157–166. https://doi.org/10.1016/j.jml.2006.03.007

van Petten, C., & Luka, B. J. (2006). Neural localization of semantic context effects in electromagnetic and hemodynamic studies. *Brain and Language*, *97*(3), 279–293. https://doi.org/10.1016/j.bandl.2005.11.003

van Petten, C., & Rheinfelder, H. (1995). Conceptual relationships between spoken words and environmental sounds: Event-related brain potential measures. *Neuropsychologia*, *33*(4), 485–508. https://doi.org/10.1016/0028-3932(94)00133-A

Vasishth, S. (2020, June 14). *Simulating data using a Latin-square design.* YouTube. https://www.youtube.com/watch?v=vpUR4qgm3jw

Vasishth, S., & Drenhaus, H. (2011). Locality in German. *Dialogue & Discourse*, *2*(1), 59–82. https://doi.org/10.5087/dad.2011.104

Vasishth, S., & Gelman, A. (2021). How to embrace variation and accept uncertainty in linguistic and psycholinguistic data analysis. *Linguistics*, *59*(5), 1311–1342. https://doi.org/10.1515/ling-2019-0051

Vasishth, S., & Nicenboim, B. (2016). Statistical Methods for Linguistic Research: Foundational Ideas – Part I. *Language and Linguistics Compass*, *10*(8), 349–369. https://doi.org/10.1111/lnc3.12201

Vasishth, S., Yadav, H., Schad, D. J., & Nicenboim, B. (2023). Sample Size Determination for Bayesian Hierarchical Models Commonly Used in Psycholinguistics. *Computational Brain & Behavior*, *6*(1), 102–126. https://doi.org/10.1007/s42113-021-00125-y

Villata, S., Tabor, W., & Franck, J. (2018). Encoding and Retrieval Interference in Sentence Comprehension: Evidence from Agreement. *Frontiers in Psychology*, *9*, 2. https://doi.org/10.3389/fpsyg.2018.00002

Vuorre, M. (2017, March 21). *Bayes Factors with Brms*. https://vuorre.com/posts/2017-03-21-bayes-factors-with-brms/

Wagenmakers, E.-J., Lodewyckx, T., Kuriyal, H., & Grasman, R. (2010). Bayesian hypothesis testing for psychologists: A tutorial on the Savage-Dickey method. *Cognitive Psychology*, *60*(3), 158–189. https://doi.org/10.1016/j.cogpsych.2009.12.001

Walsh, B. (2004). *Markov Chain Monte Carlo and Gibbs Sampling*. *Lecture Notes for EEB 581, Version 26, April*. https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=d95f3929cafa0514dc6d16e4be9c20538caf344f

Ward, P., & Sturt, P. (2007). Linguistic focus and memory: An eye movement study. *Memory & Cognition*, *35*(1), 73–86. https://doi.org/10.3758/BF03195944

Yadav, H., Smith, G., Reich, S., & Vasishth, S. (2023). Number feature distortion modulates cue-based retrieval in reading. *Journal of Memory and Language*, *129*, 104400. https://doi.org/10.1016/j.jml.2022.104400

Yadav, H., Smith, G., & Vasishth, S. (2021). *Feature encoding modulates cue-based retrieval: Modeling interference effects in both grammatical and ungrammatical sentences.* https://doi.org/10.31234/osf.io/76aex

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